TrustForge™

Complete System Screenshots & Feature Verification Guide

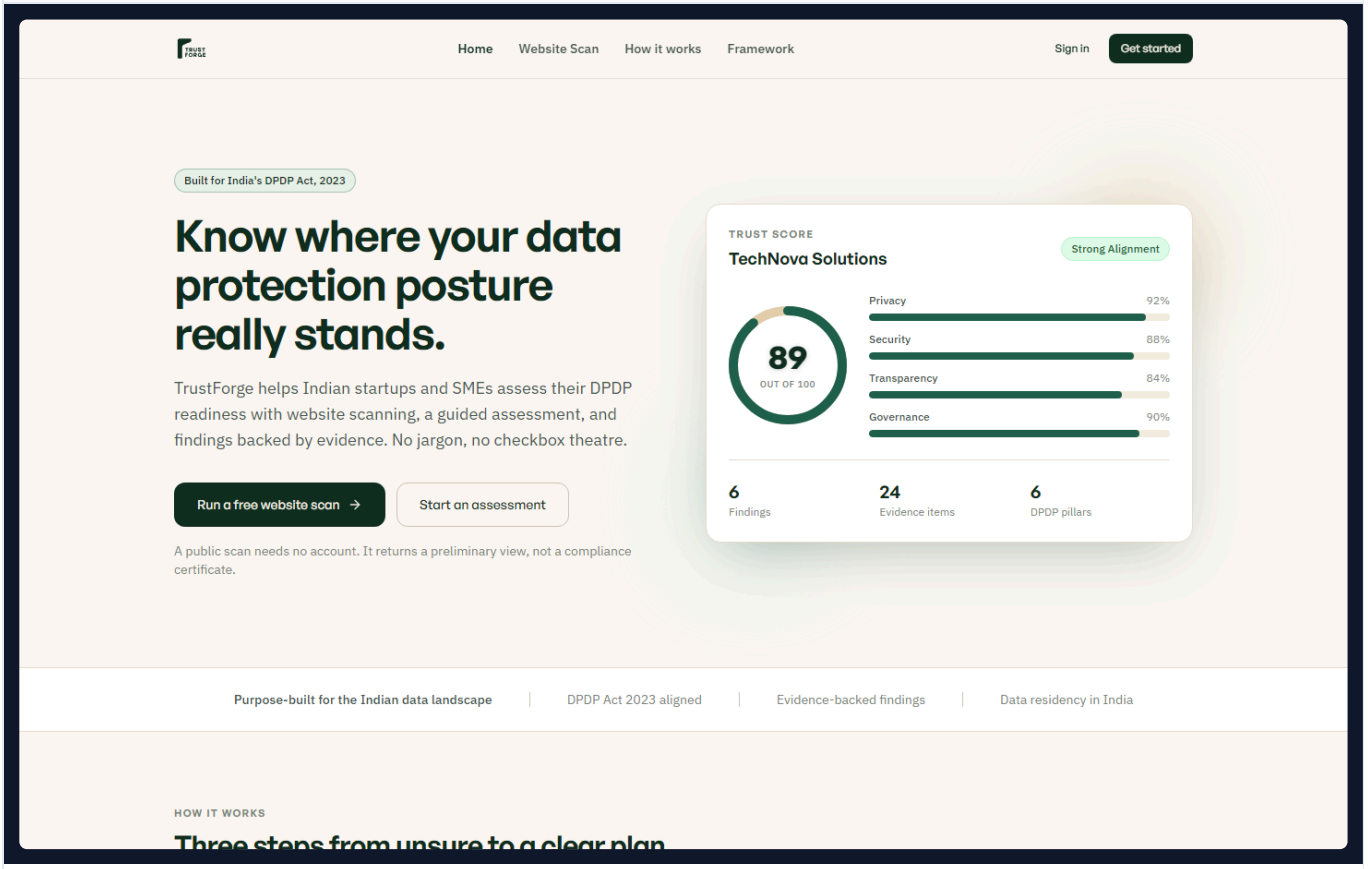
Platform Version:	`v1.0.0-release (Golden Master)`
Verified Commit SHA:	`035429f6c76d7ab6d14f2053b3e9d2050f493672`
Statutory Foundation:	`DPDP Act 2023 (Act No. 22 of 2023, India)`
Authentication & Tokens:	`PBKDF2 (SHA-512, 10k iter), HMAC-SHA256 (tf_tok_)`
Trust Score Structure:	`6 Canonical Pillars (C1-C10 Mappings, min(TS, 40) Caps)`
Benchmarking & TOPSIS:	`10 Candidate Slots (M01-M10), 7 Primary Criteria, $0 Spend`

Executive Architecture Summary: TrustForge is an automated compliance auditing and model benchmarking platform designed to evaluate enterprise data privacy posture against India's Digital Personal Data Protection (DPDP) Act 2023. This document delivers complete visual evidence of all customer-facing workflows, the complete 14-step statutory assessment wizard, empirical analytics suites (hybrid RAG with Reciprocal Rank Fusion and TOPSIS MCDM), and the isolated Administrative Governance Console. Every figure represents active application execution verified under strict continuous integration and zero-paid commercial inference boundary enforcement.

Customer Platform & Identity Workflows

Figure 1: TrustForge Public Landing Page & Privacy Assurance Hero

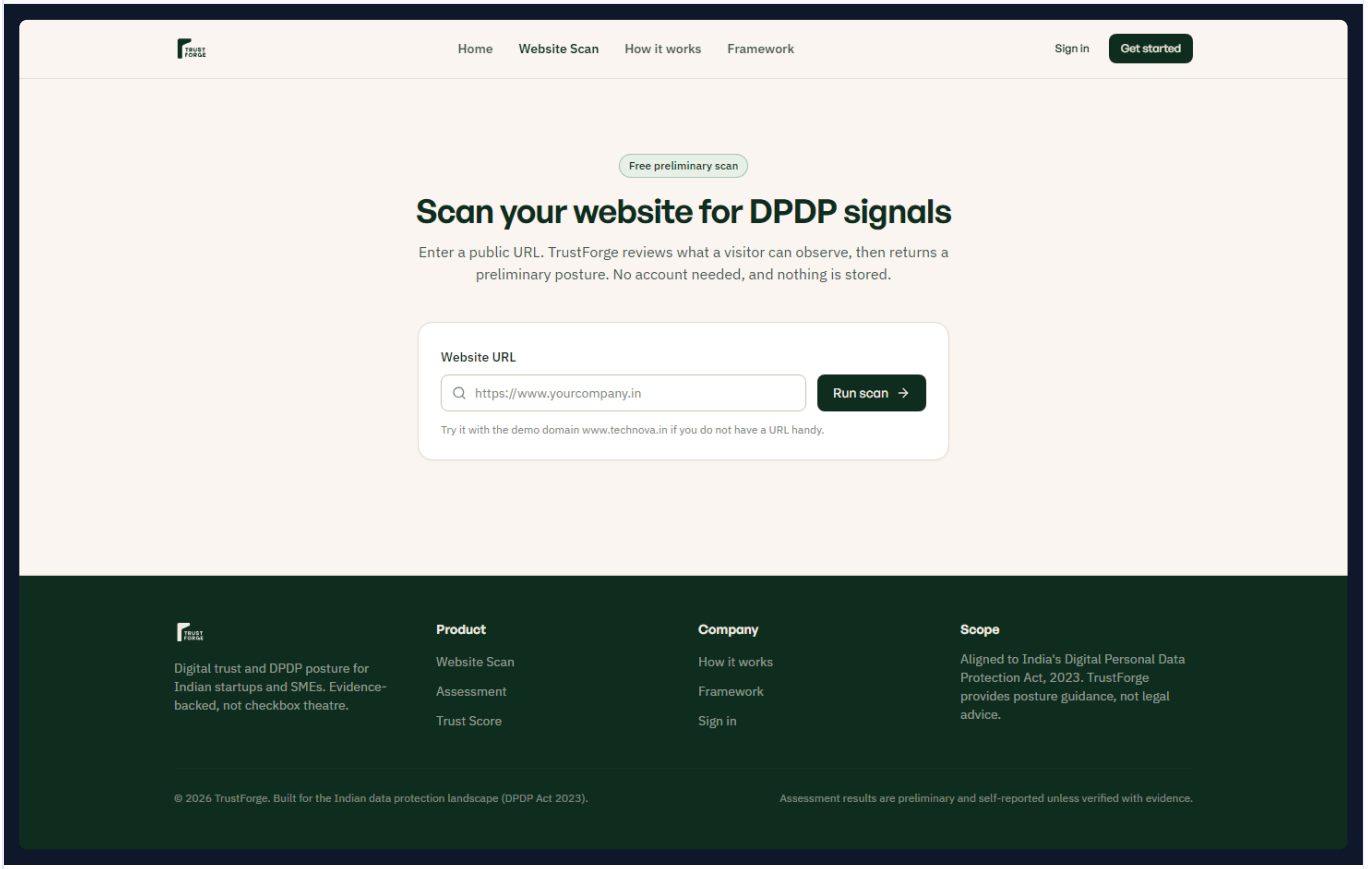
`Route: / (Public)` Role: Public Visitor



The landing page provides the primary interface for organizations seeking automated self-assessment against the DPDP Act 2023. It highlights the platform's core architectural components: hybrid RAG retrieval, multi-model benchmarking, and evidence management. Visitors can initiate a non-authenticated preliminary domain scan or proceed to create an enterprise account.

Figure 2: Instant Public Domain Assessment & Scanner

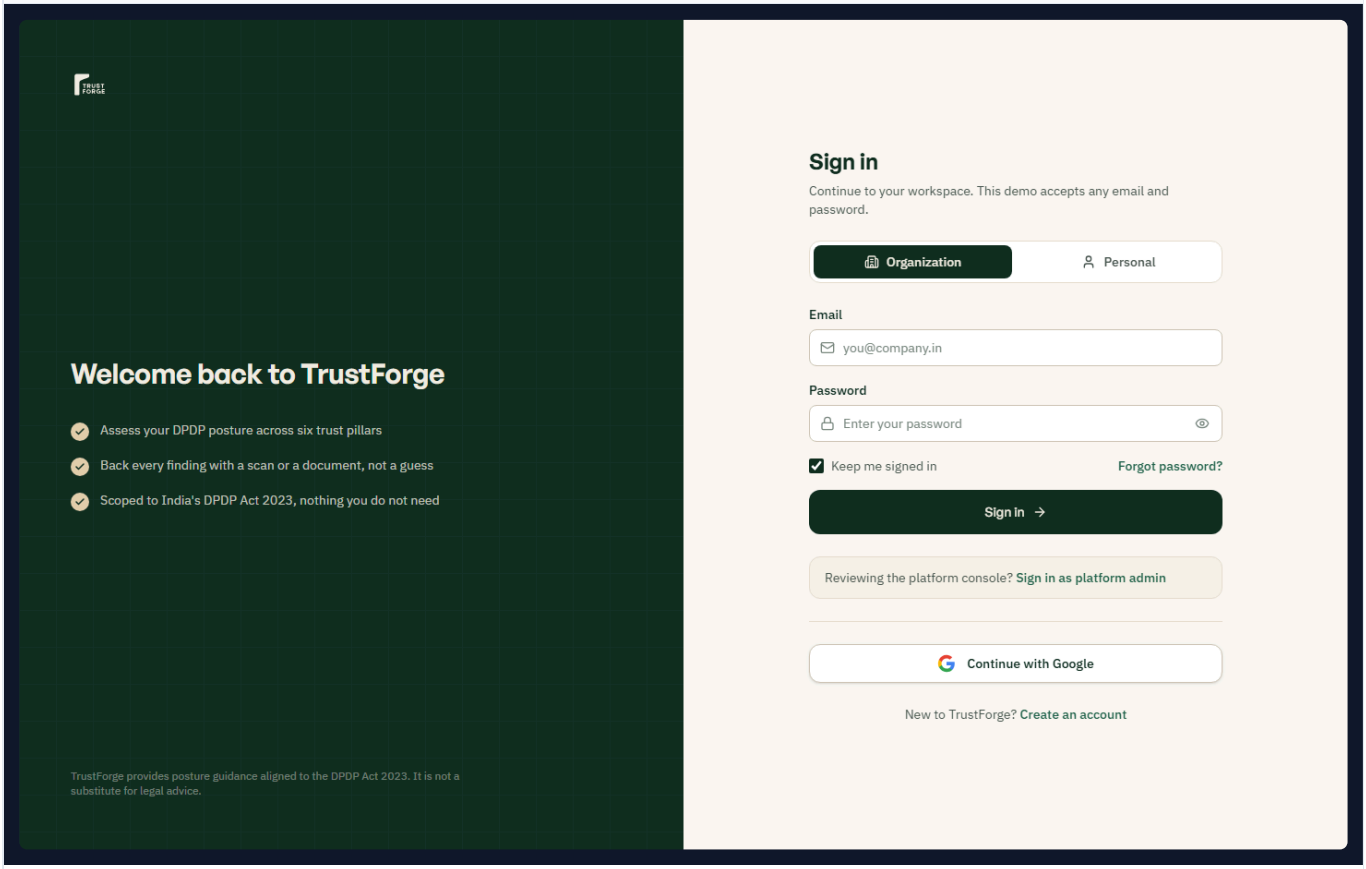
`Route: /scan (Public)` Role: Public Visitor



Executes an unauthenticated, non-invasive perimeter scan of a target web domain to check public privacy configurations. It checks for the presence of cookie banners, privacy policy links, SSL/TLS transport availability, and detectable third-party scripts. The resulting summary identifies observable perimeter issues and provides recommendations for full compliance auditing.

Figure 3: Customer Authentication Portal

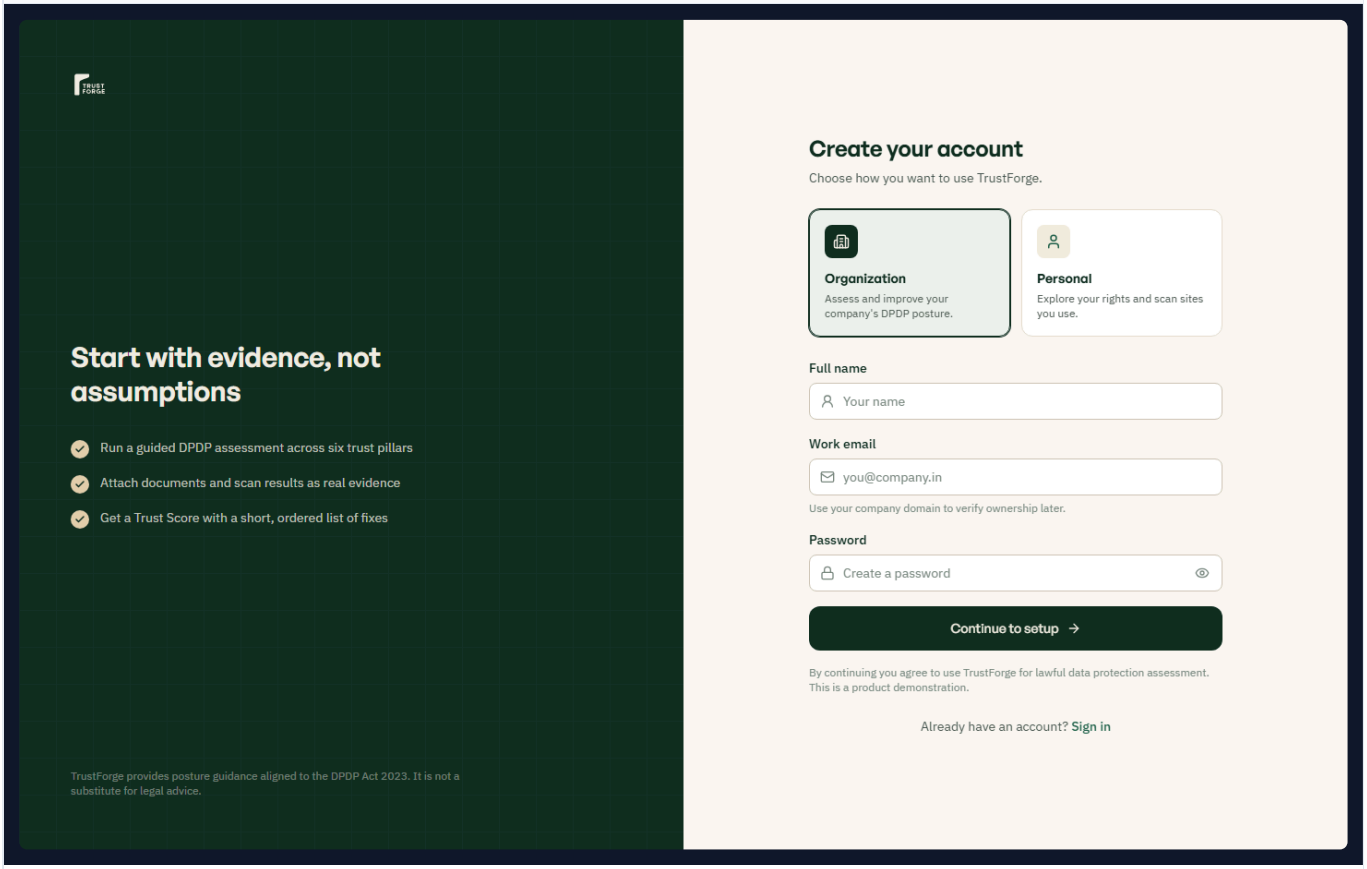
`Route: /login (Public/Client)` Role: Client User / Compliance Lead



Implements multi-tenant credential authentication using standard PBKDF2 password hashing (SHA-512, 10,000 iterations) with unique salt. Upon successful authentication, the backend issues an HMAC-SHA256 session token with tenant organization claims. Tenant-scoped route middleware ensures users cannot access or view data belonging to other registered organizations.

Figure 4: Enterprise Registration & Account Creation

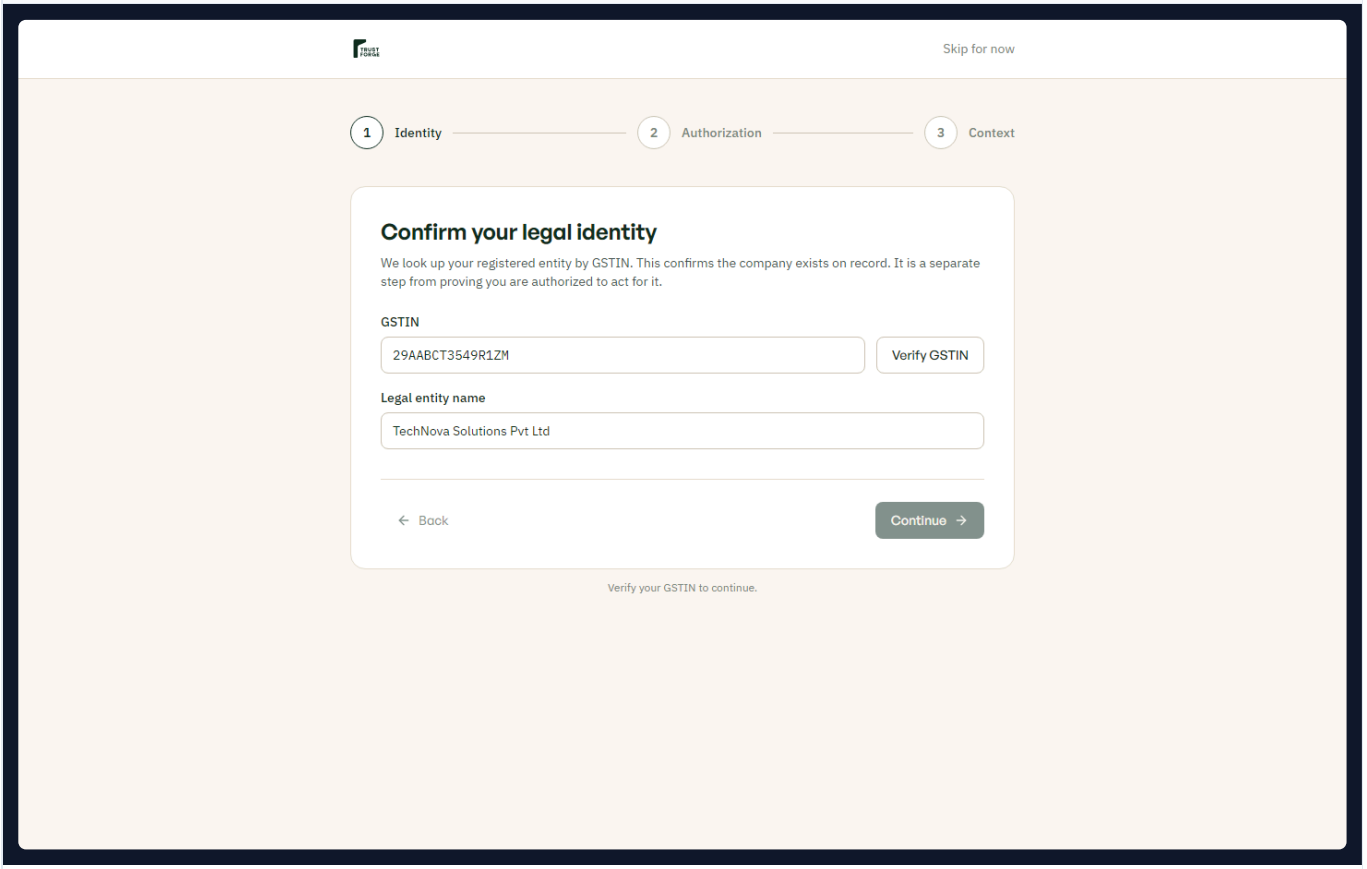
`Route: /signup (Public)` Role: Prospective Client



Allows new enterprise compliance personnel to register a tenant account with email validation and password complexity checks. Generates a dedicated organization identifier and provisions initial tenant records in the durable application store. Upon registration, the user is transitioned into the onboarding workflow to configure initial organizational parameters.

Figure 5: Enterprise Organization Profile Setup Wizard

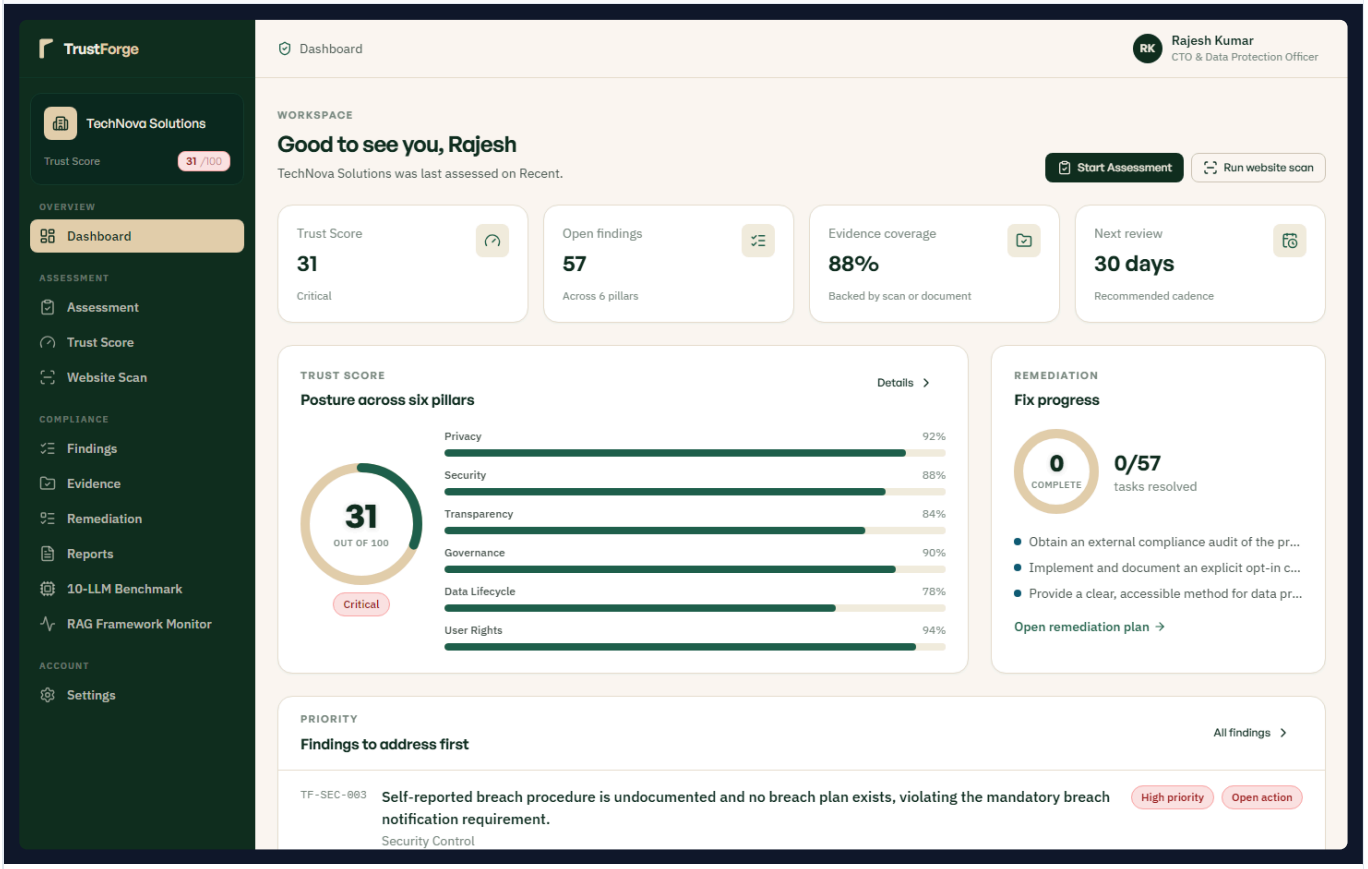
`Route: /onboarding (Authenticated Client)` Role: Organization Administrator



Collects organizational metadata including corporate entity name, industry sector, primary website domain, and designated DPO details. Determines whether the entity self-classifies as a Significant Data Fiduciary (SDF) under DPDP Act Section 10 obligations. Initializes default assessment baselines and generates domain verification tokens for ownership validation.

Figure 6: Executive Privacy Compliance Dashboard

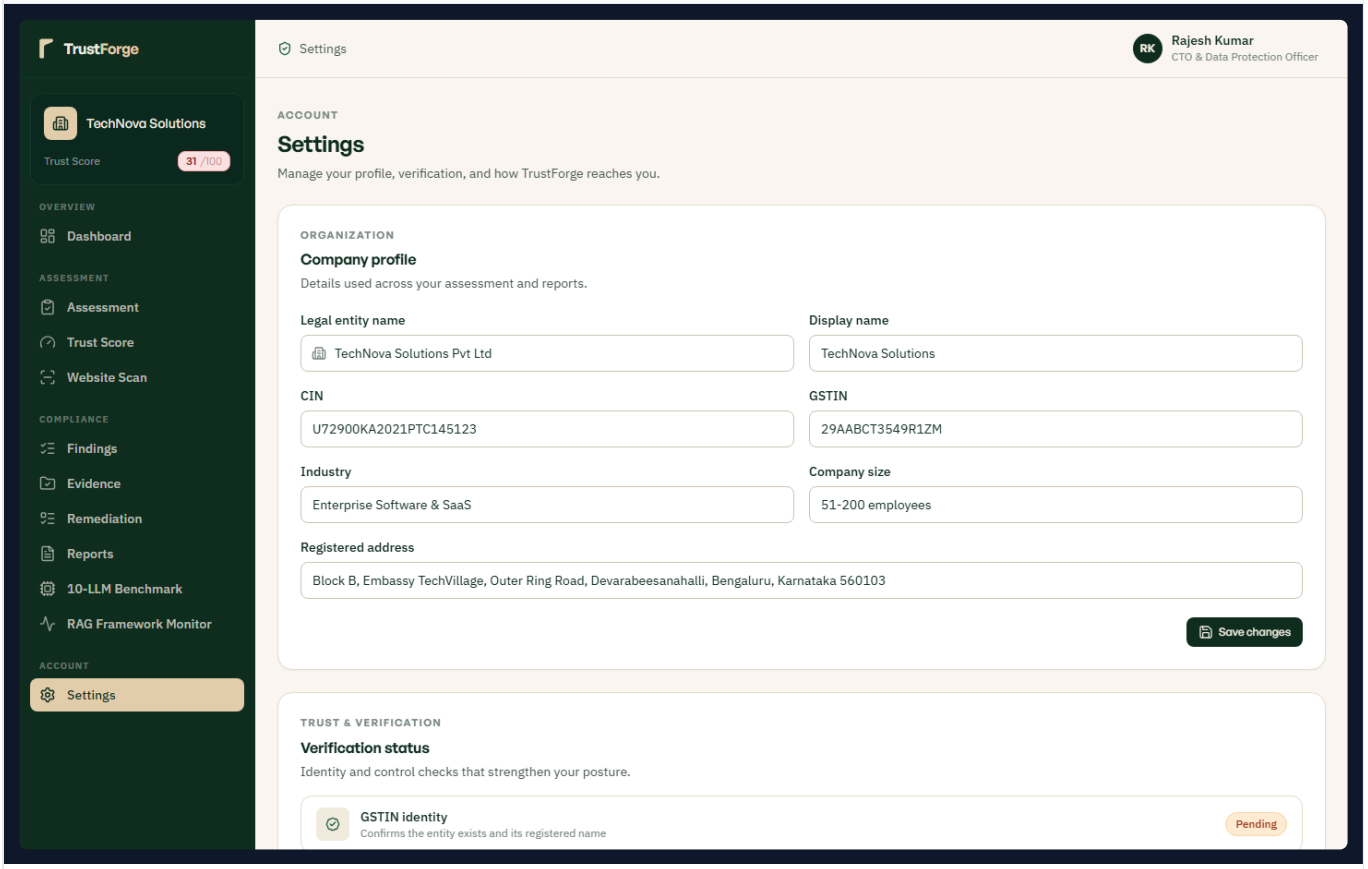
`Route: /dashboard (Authenticated Client)` Role: Compliance Officer / DPO



Displays the tenant's current Trust Score, assessment completion progress, active findings count, and evidence metrics. Summary cards categorize compliance observations across the six statutory pillars and highlight open remediation items. Provides direct links to start or resume the 14-step assessment, trigger perimeter scans, and export summary reports.

Figure 7: Organization Profile & Data Fiduciary Configuration

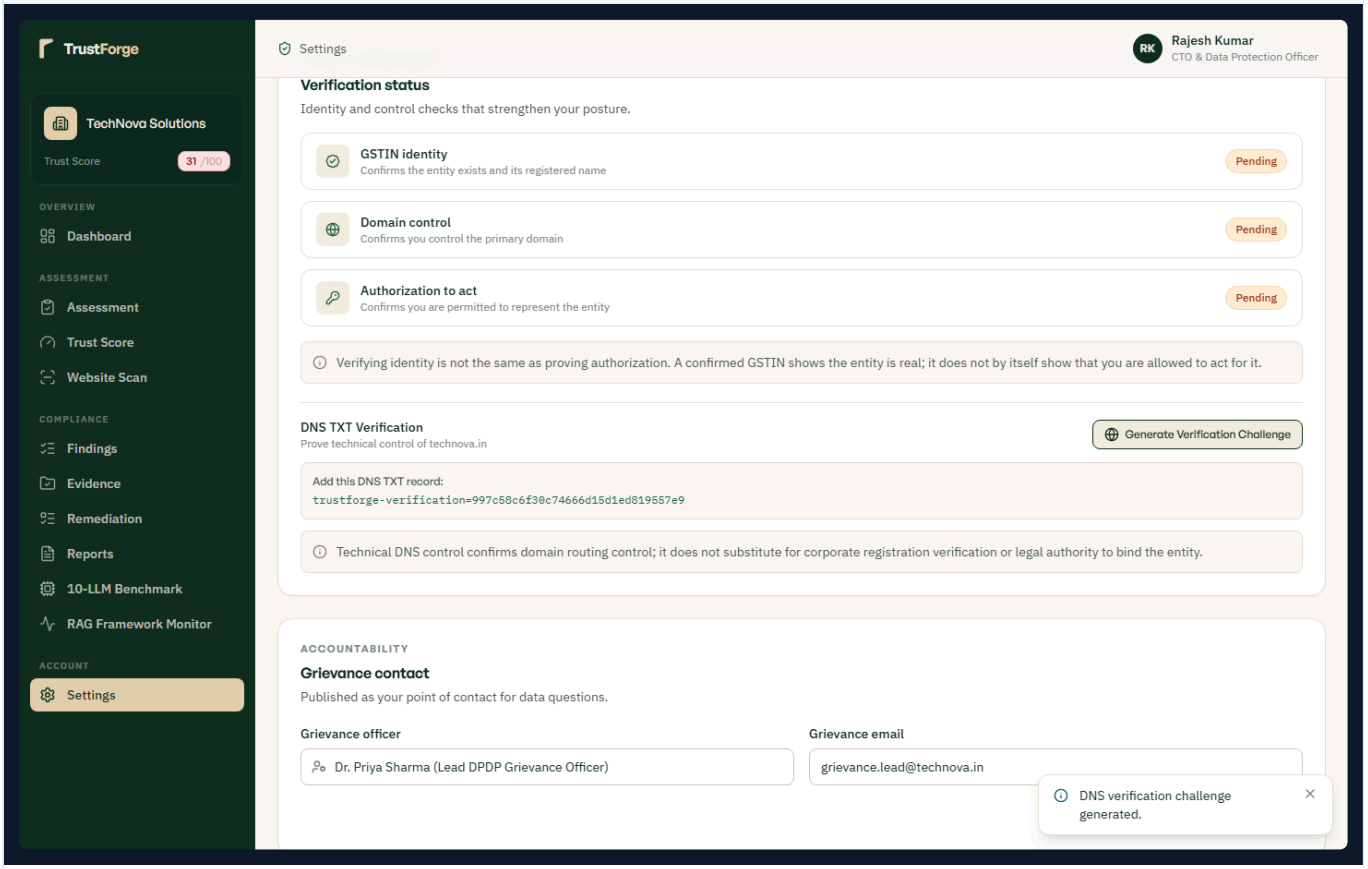
`Route: /settings/organization (Client Admin)` Role: Organization Administrator



Allows administrators to update organizational records, designated Data Protection Officer contact details, and corporate headquarters. Enables maintenance of corporate sector classifications and review of applicable DPDP Act statutory requirements. All profile updates are recorded in the organization audit history with timestamps and user identifiers.

Figure 8: Automated DNS TXT Domain Ownership Verification

`Route: /settings/organization#domain-verification` Role: Organization Administrator

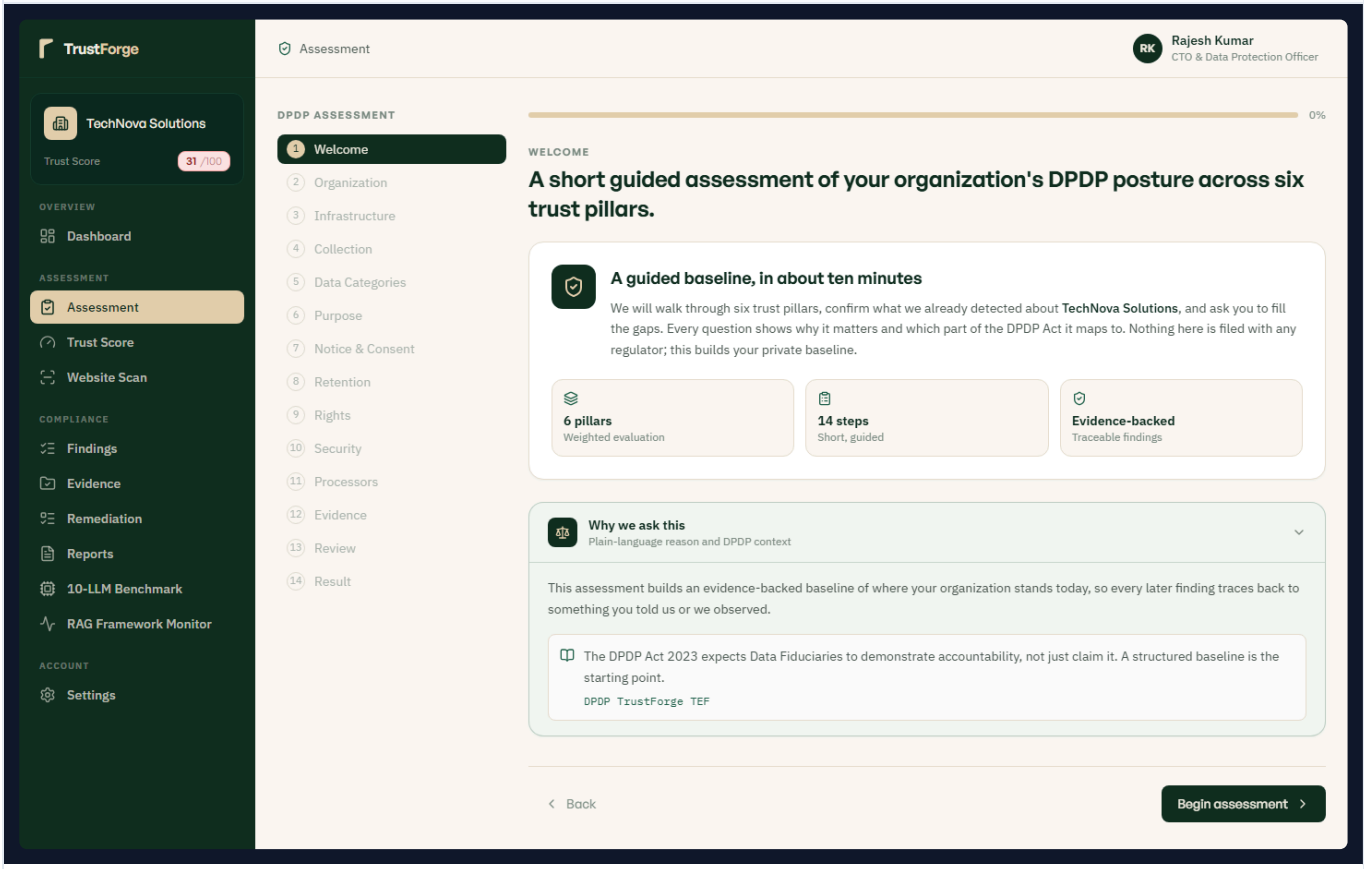


Generates a unique DNS TXT challenge token formatted as `trustforge-site-verification=` to verify domain ownership. The backend performs asynchronous DNS lookups against public nameservers to confirm token propagation prior to active scanning. Ensures website scans and compliance scoring are only initiated for domains legitimately managed by the tenant organization.

14-Step Statutory Assessment Wizard

Figure 9: Assessment Step 1 — Governance & Accountability Architecture

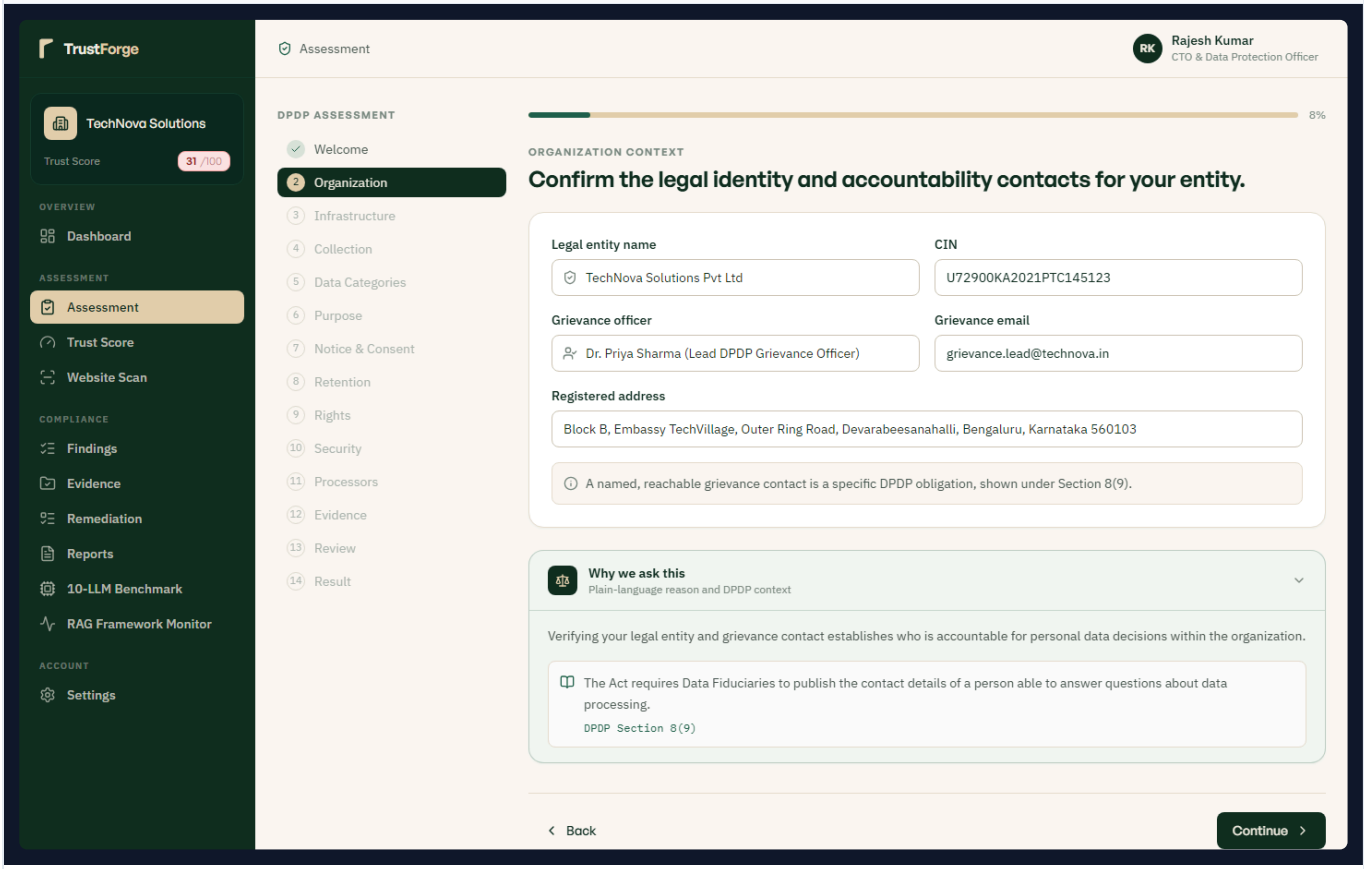
`Route: /assessment/wizard/step-1` Role: Compliance Assessor



Evaluates organizational data governance mechanisms, internal compliance reporting hierarchies, and DPO appointments. Maps directly to DPDP Act Section 8(1) and Section 10(2)(a) obligations for responsible data fiduciary governance. Audits formal documentation, board-level oversight policies, and designated privacy officer operational roles.

Figure 10: Assessment Step 2 — Consent Architecture & Validation

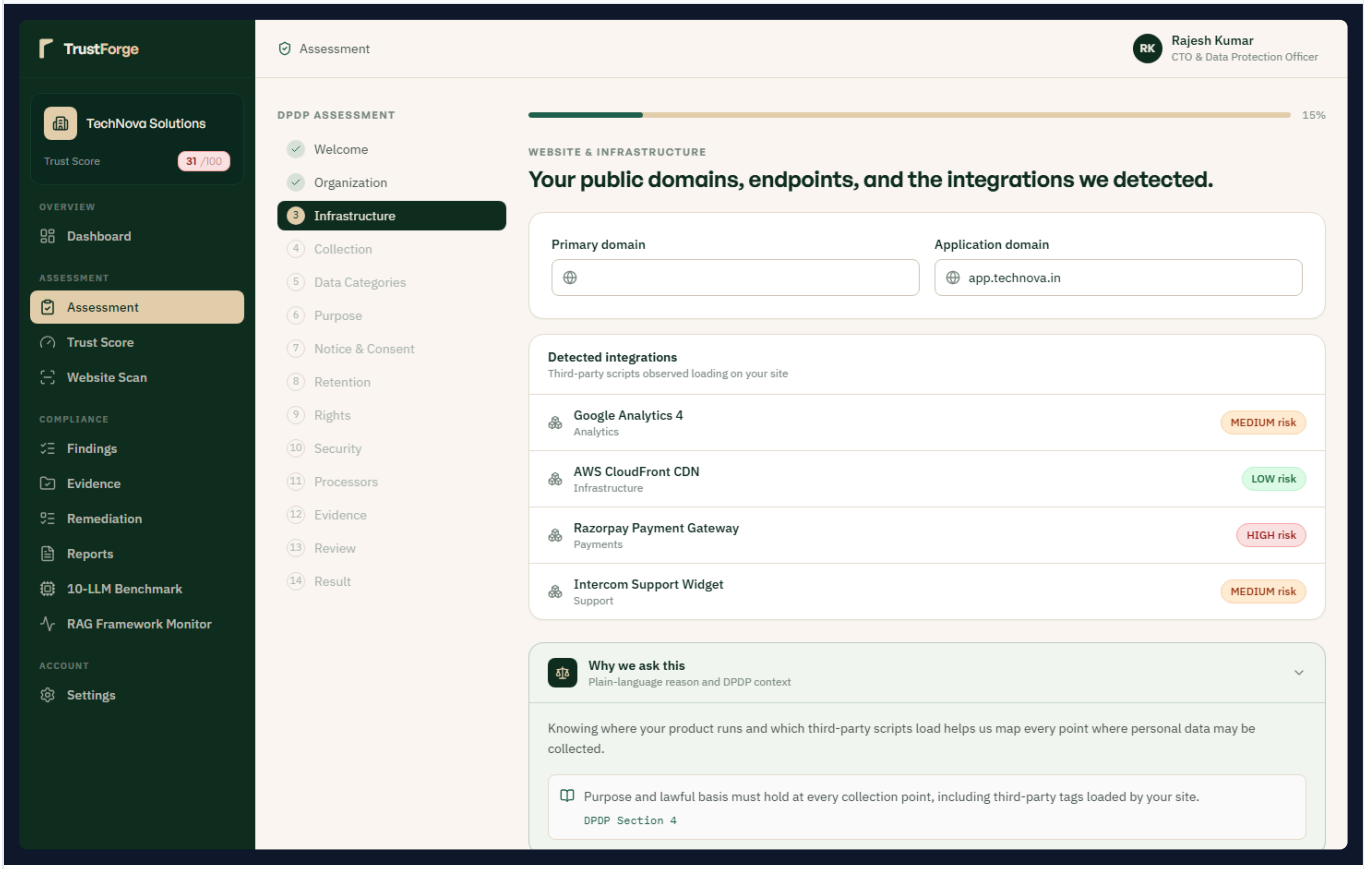
`Route: /assessment/wizard/step-2` Role: Compliance Assessor



Audits affirmative, freely given, specific, informed, and unambiguous consent mechanisms under DPDP Act Section 6(1). Checks that opt-in selections are unchecked by default, bundled terms are avoided, and withdrawal procedures are accessible. Verifies timestamped consent recordkeeping and specific purpose declarations for each data processing activity.

Figure 11: Assessment Step 3 — Multilingual Statutory Notice Delivery

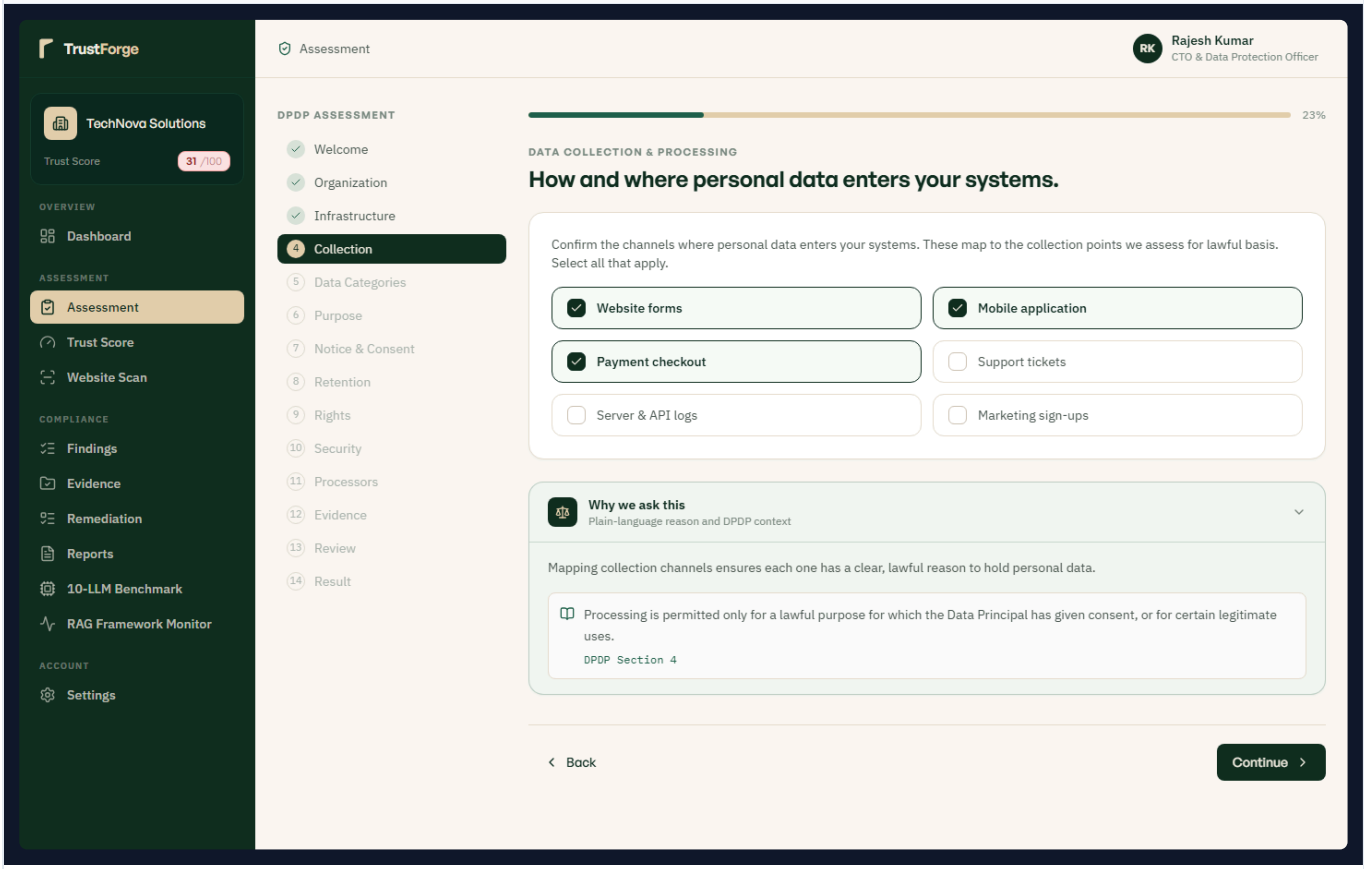
`Route: /assessment/wizard/step-3` Role: Compliance Assessor



Evaluates statutory privacy notice completeness, itemization of collected personal data, and DPO contact details under Section 5. Checks support for notice availability in English and the 22 languages specified in the Eighth Schedule of the Constitution of India. Verifies clear instructions on how data principals may exercise rights or lodge grievances with the fiduciary.

Figure 12: Assessment Step 4 — Consent Manager Integration & Interoperability

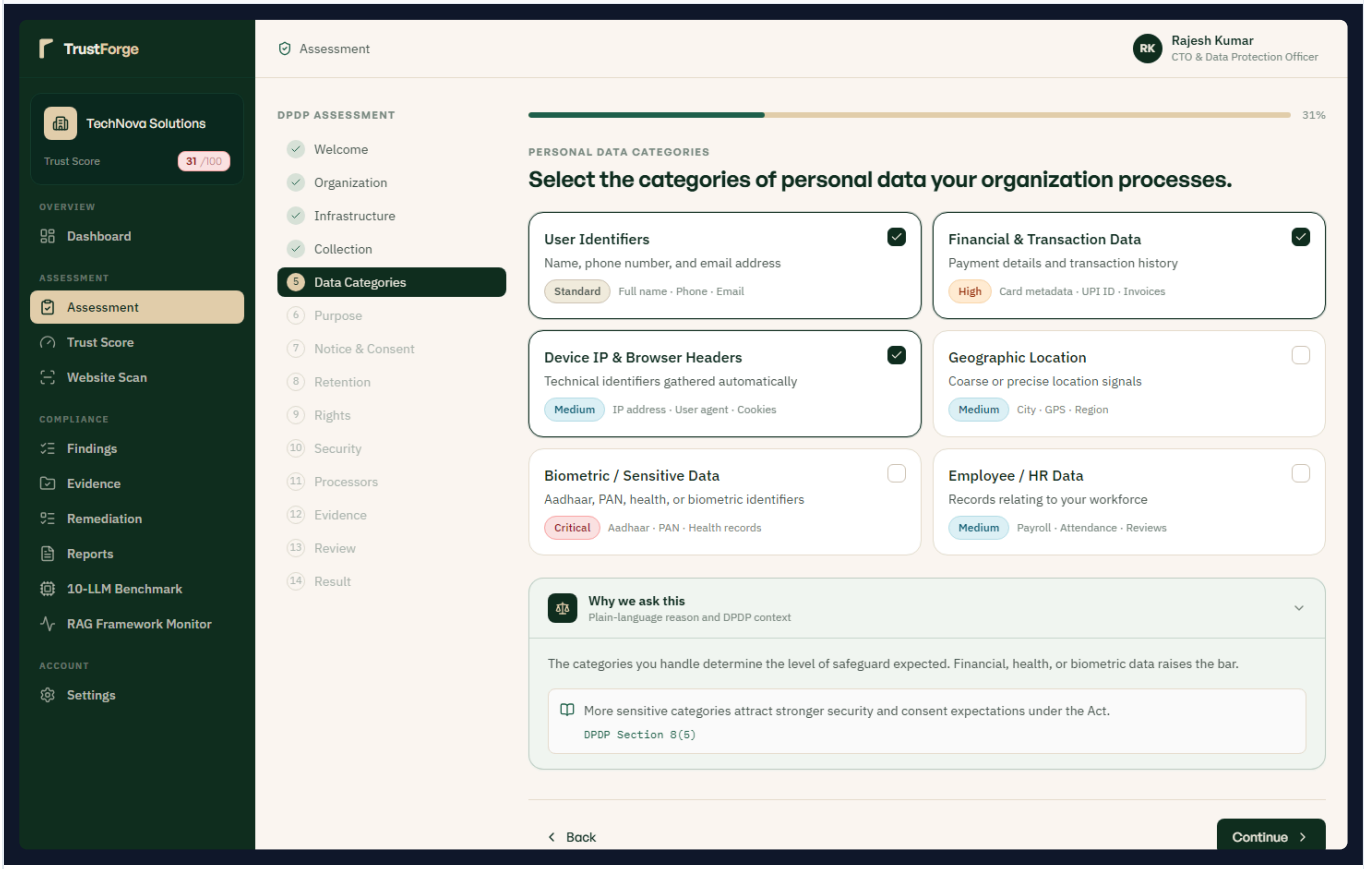
`Route: /assessment/wizard/step-4` Role: Compliance Assessor



Assesses technical readiness for integrating with Data Protection Board-registered Consent Managers under Section 6(7)-(9). Evaluates API interfaces and workflows enabling data principals to view, manage, and withdraw consent through unified mechanisms. Audits procedures for propagating consent revocation across internal databases and third-party data processors.

Figure 13: Assessment Step 5 — Purpose Limitation & Data Minimization

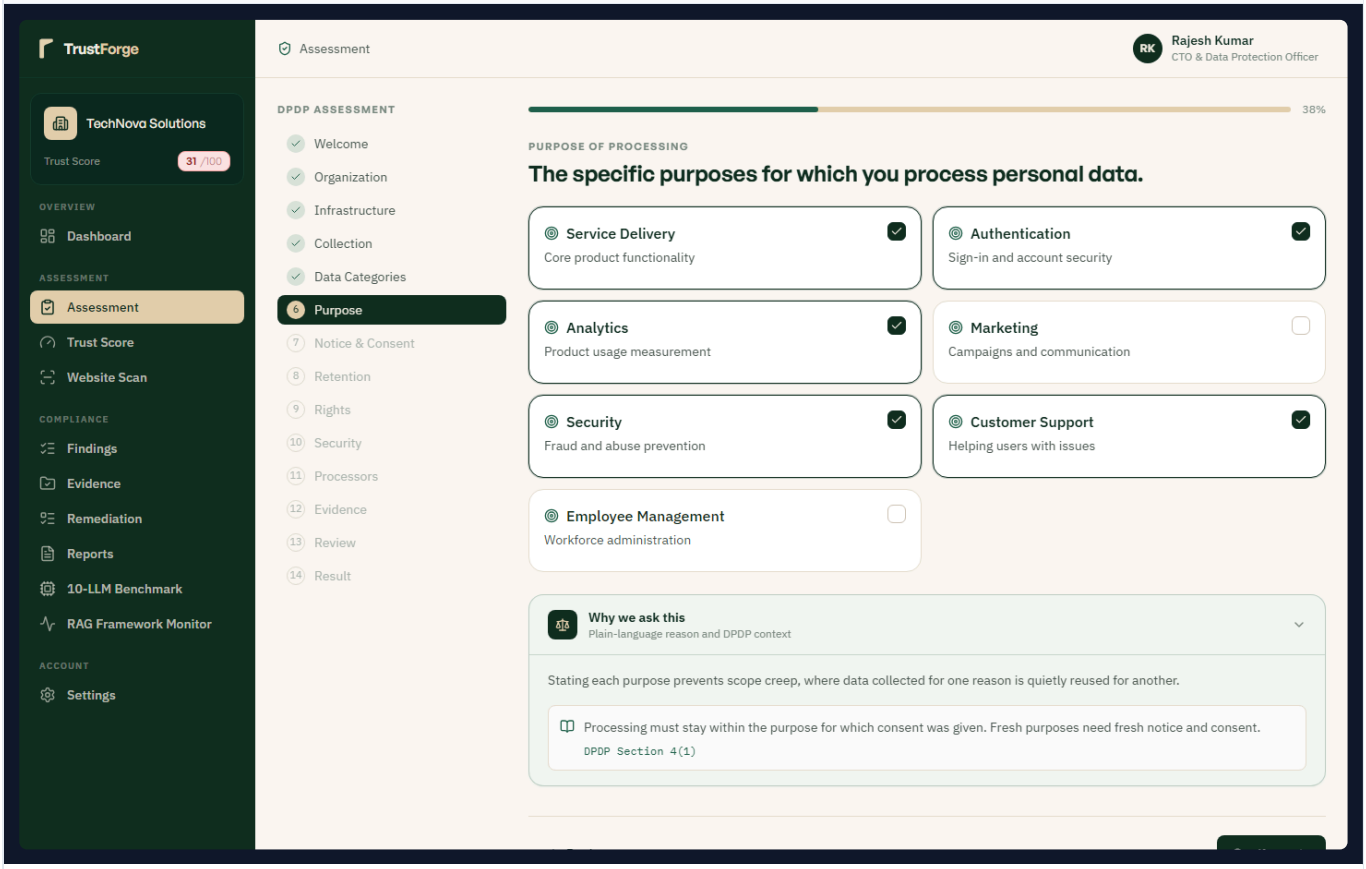
`Route: /assessment/wizard/step-5` Role: Compliance Assessor



Assesses adherence to statutory collection boundaries under DPDP Act Section 4 and Section 7, ensuring only necessary data is collected. Identifies potential collection creep, collateral data gathering, and secondary use beyond the consented purpose. Checks safeguards preventing personal data collected for transactional purposes from being reused for unauthorized profiling.

Figure 14: Assessment Step 6 — Storage Limitation & Secure Erasure

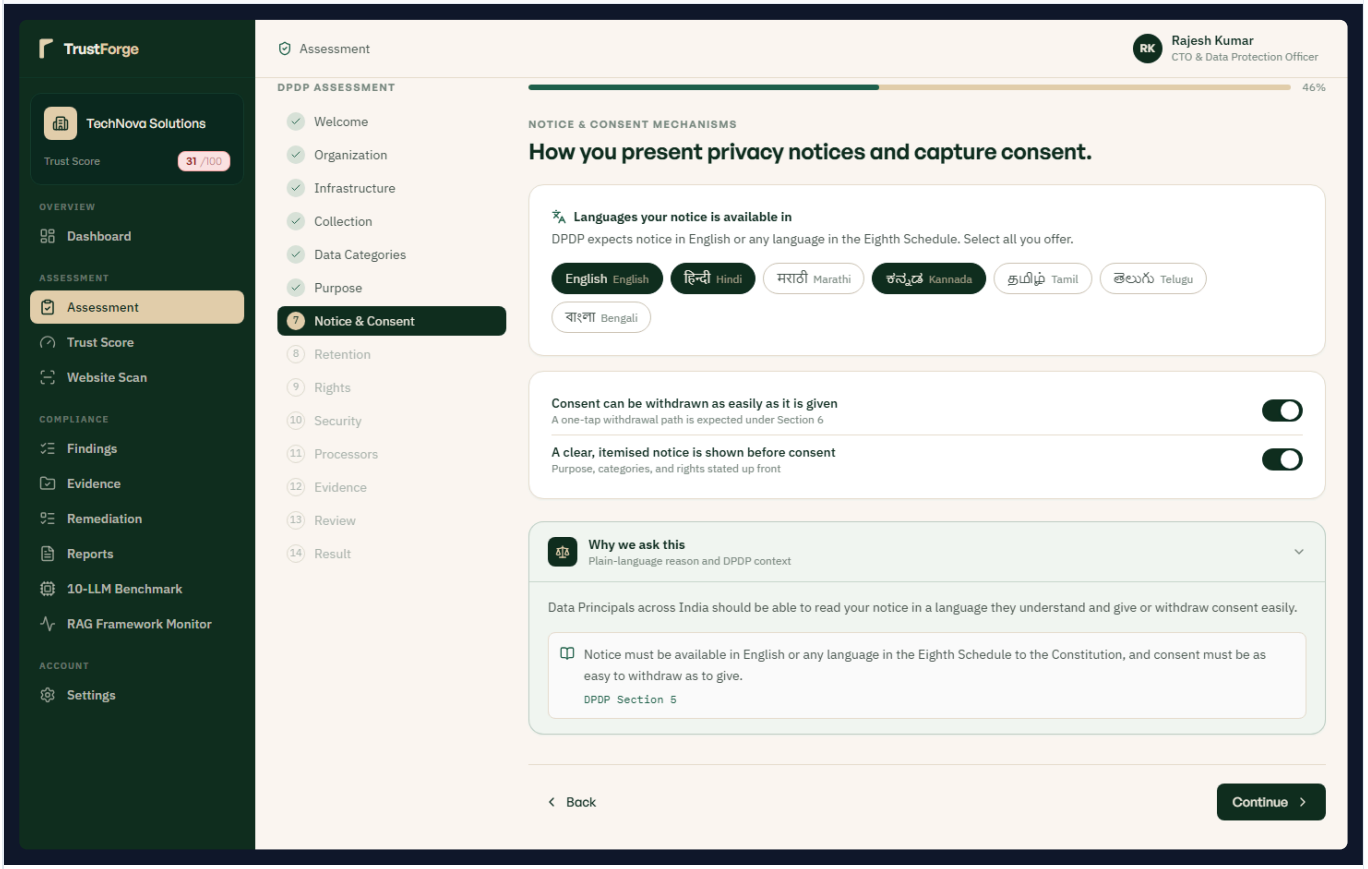
`Route: /assessment/wizard/step-6` Role: Compliance Assessor



Evaluates defined data retention schedules, automated archival triggers, and secure deletion protocols under Section 8(7). Verifies that personal data is deleted once the specified processing purpose is accomplished or consent is formally revoked. Audits tombstone tracking records, deletion verification logs, and legal hold override controls.

Figure 15: Assessment Step 7 — Data Quality, Completeness & Accuracy

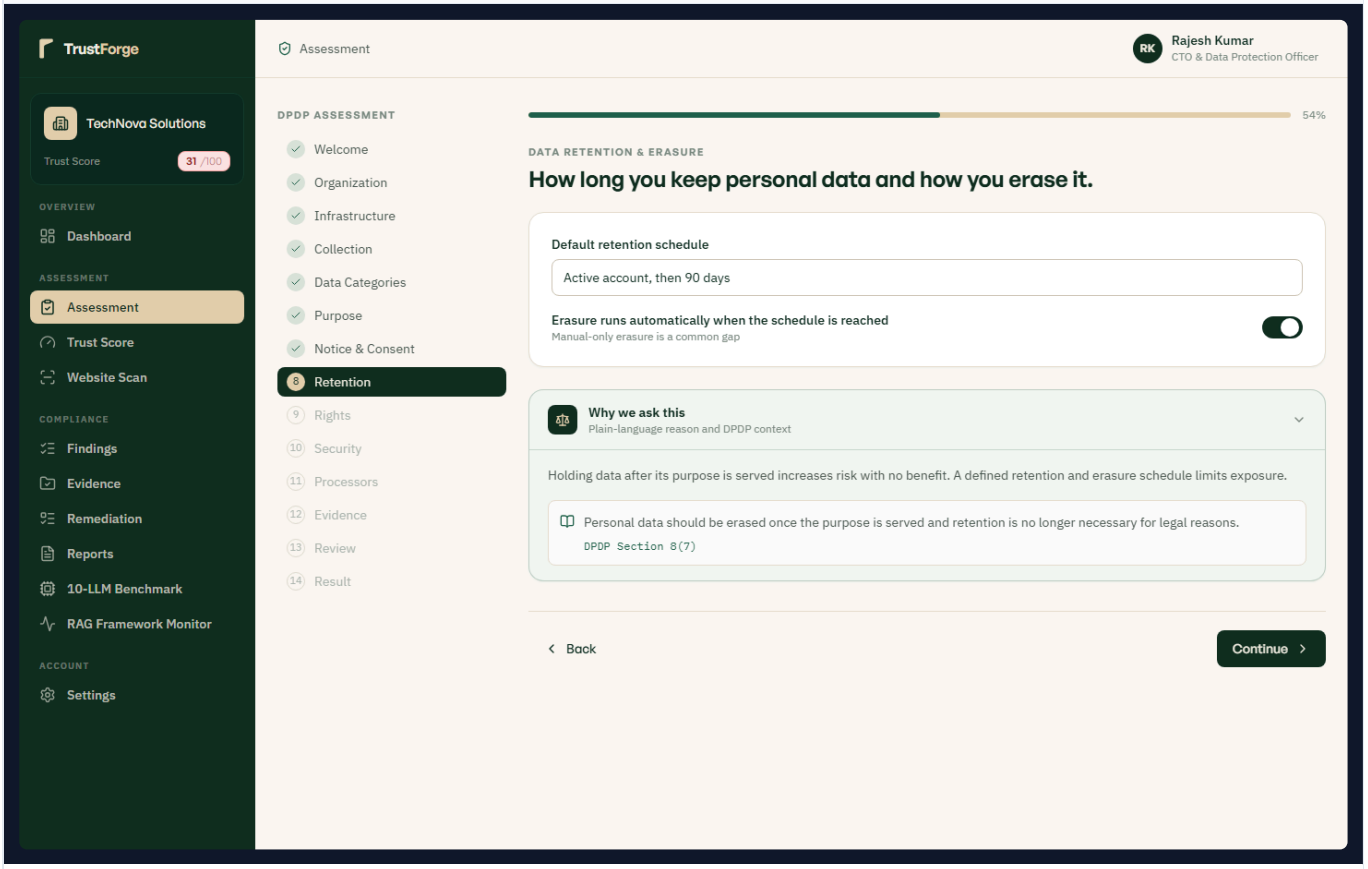
`Route: /assessment/wizard/step-7` Role: Compliance Assessor



Assesses mechanisms for maintaining data accuracy and completeness when personal data is used to make decisions affecting data principals. Maps directly to DPDP Act Section 8(3) obligations requiring reasonable measures to ensure data consistency. Reviews user self-service correction interfaces, synchronization processes, and downstream processor notification workflows.

Figure 16: Assessment Step 8 — Reasonable Technical & Security Safeguards

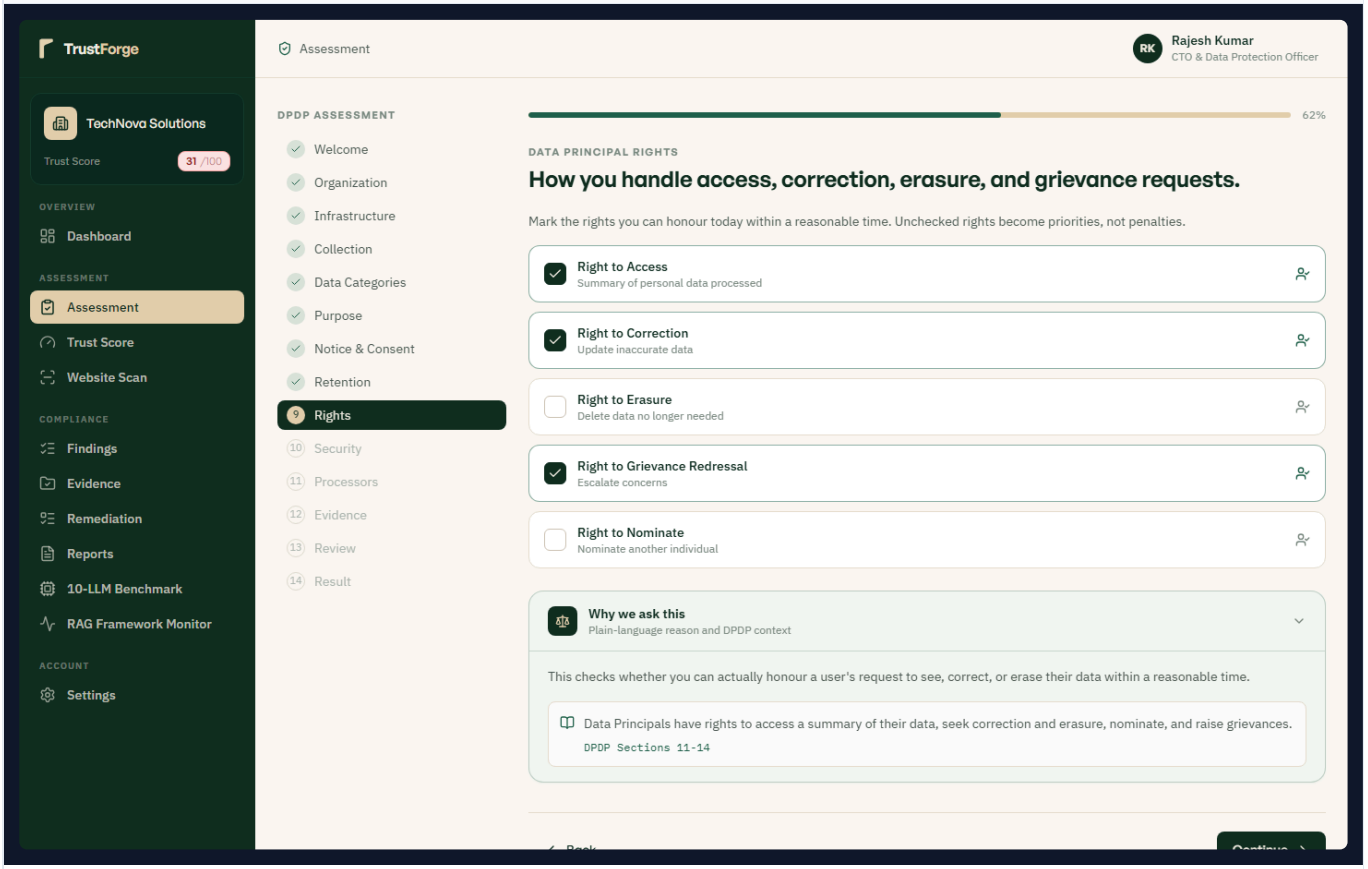
`Route: /assessment/wizard/step-8` Role: Compliance Assessor



Audits technical security controls including TLS encryption in transit, role-based access control, and credential safeguards. Evaluates organizational adherence to DPDP Act Section 8(5) requiring reasonable security safeguards to prevent data breaches. Reviews vulnerability review cadences, system patching policies, and multi-factor authentication enforcement.

Figure 17: Assessment Step 9 — Personal Data Breach Incident Management

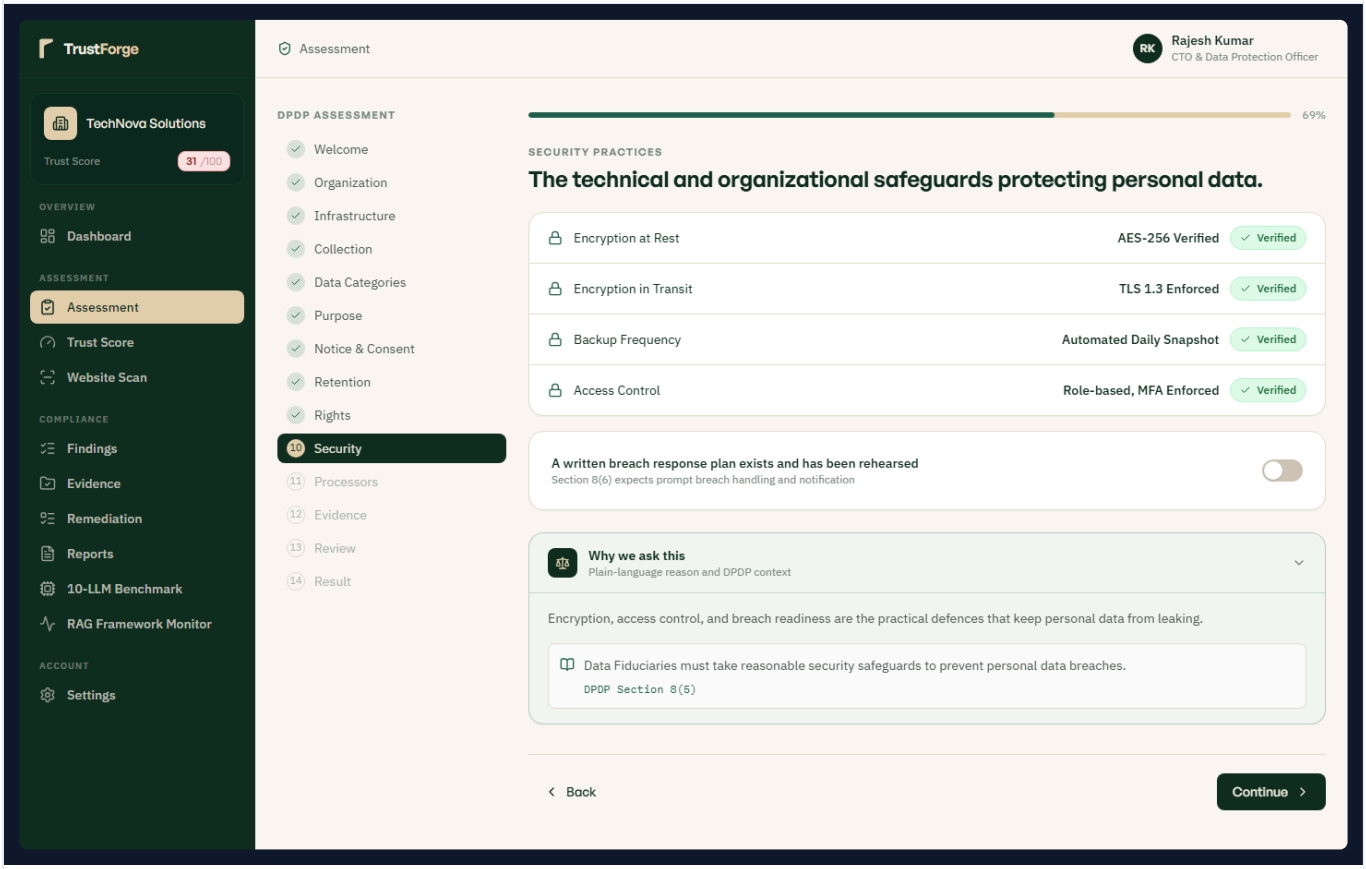
`Route: /assessment/wizard/step-9` Role: Compliance Assessor



Reviews incident response procedures and formal breach reporting plans as required under DPDP Act Section 8(6). Evaluates playbooks for notifying the Data Protection Board of India and affected data principals upon discovering a personal data breach. Checks incident containment steps, forensic audit logging mechanisms, and statutory reporting escalation timelines.

Figure 18: Assessment Step 10 — Children's Data & Persons with Disabilities

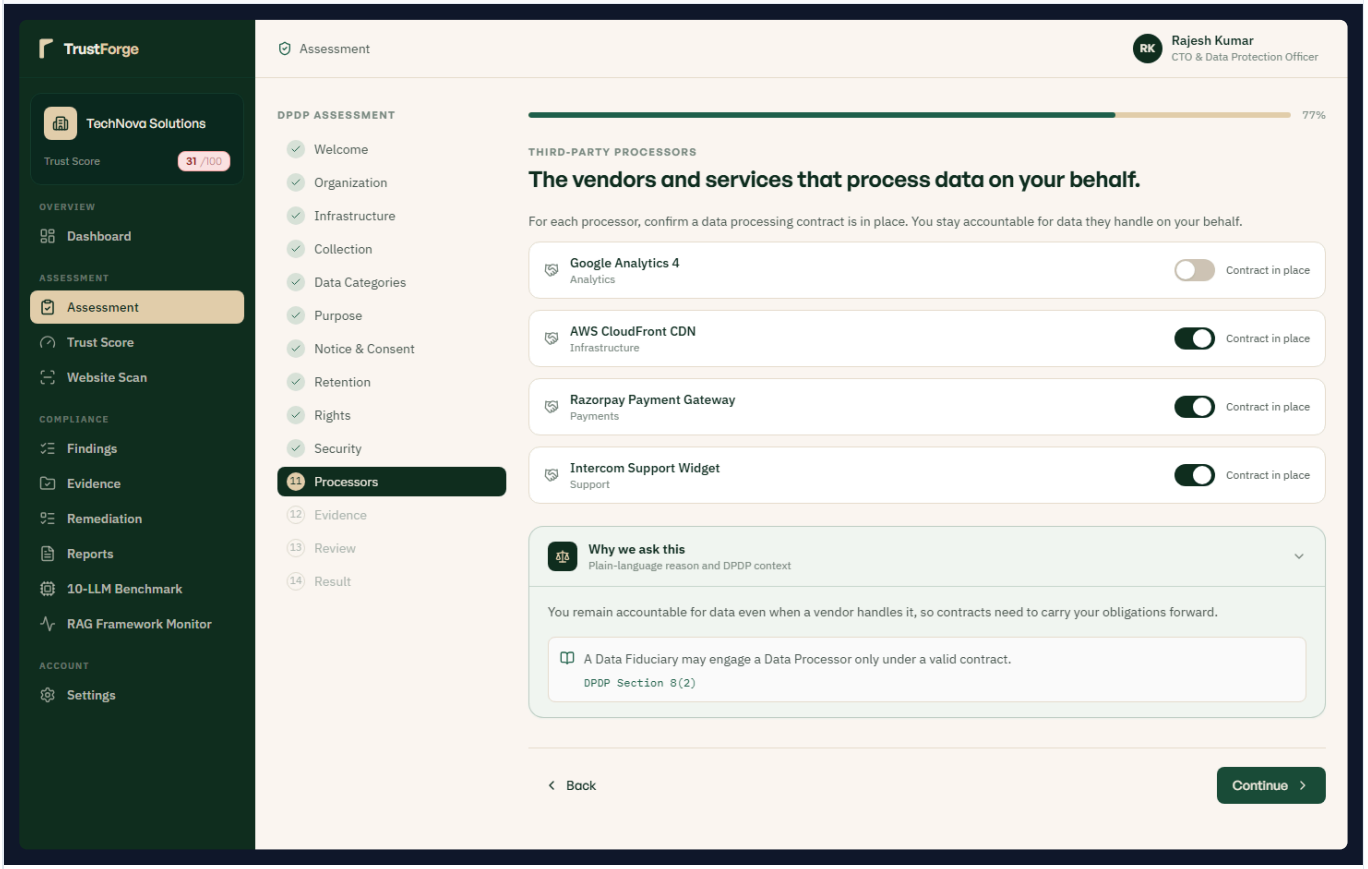
`Route: /assessment/wizard/step-10` Role: Compliance Assessor



Evaluates parental consent verification protocols and strict prohibitions against tracking, profiling, or targeted advertising to minors. Enforces compliance with DPDP Act Section 9 mandates protecting children and persons under legal guardianship. Reviews age-verification workflows, neutral gating methods, and absence of harmful behavioral monitoring practices.

Figure 19: Assessment Step 11 — Data Principal Rights Redressal Architecture

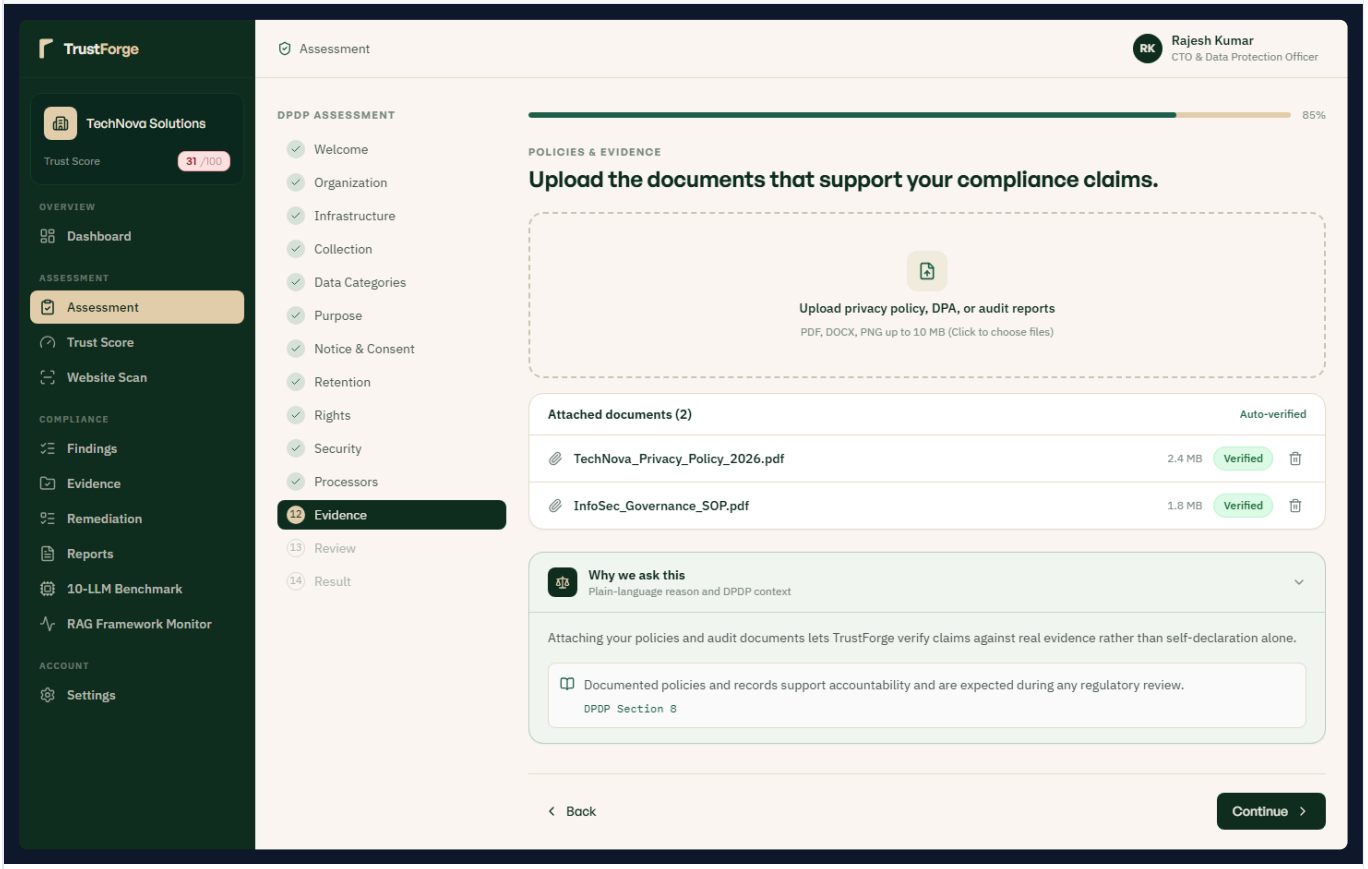
`Route: /assessment/wizard/step-11` Role: Compliance Assessor



Audits self-service procedures for processing data principal requests: right to access (Sec 11), correction (Sec 12), and erasure. Evaluates identity verification methods, data export formats, and response tracking against statutory fulfillment windows. Reviews mechanisms supporting the right of nomination in the event of death or legal incapacity under Section 14.

Figure 20: Assessment Step 12 — Grievance Redressal Mechanism & Timelines

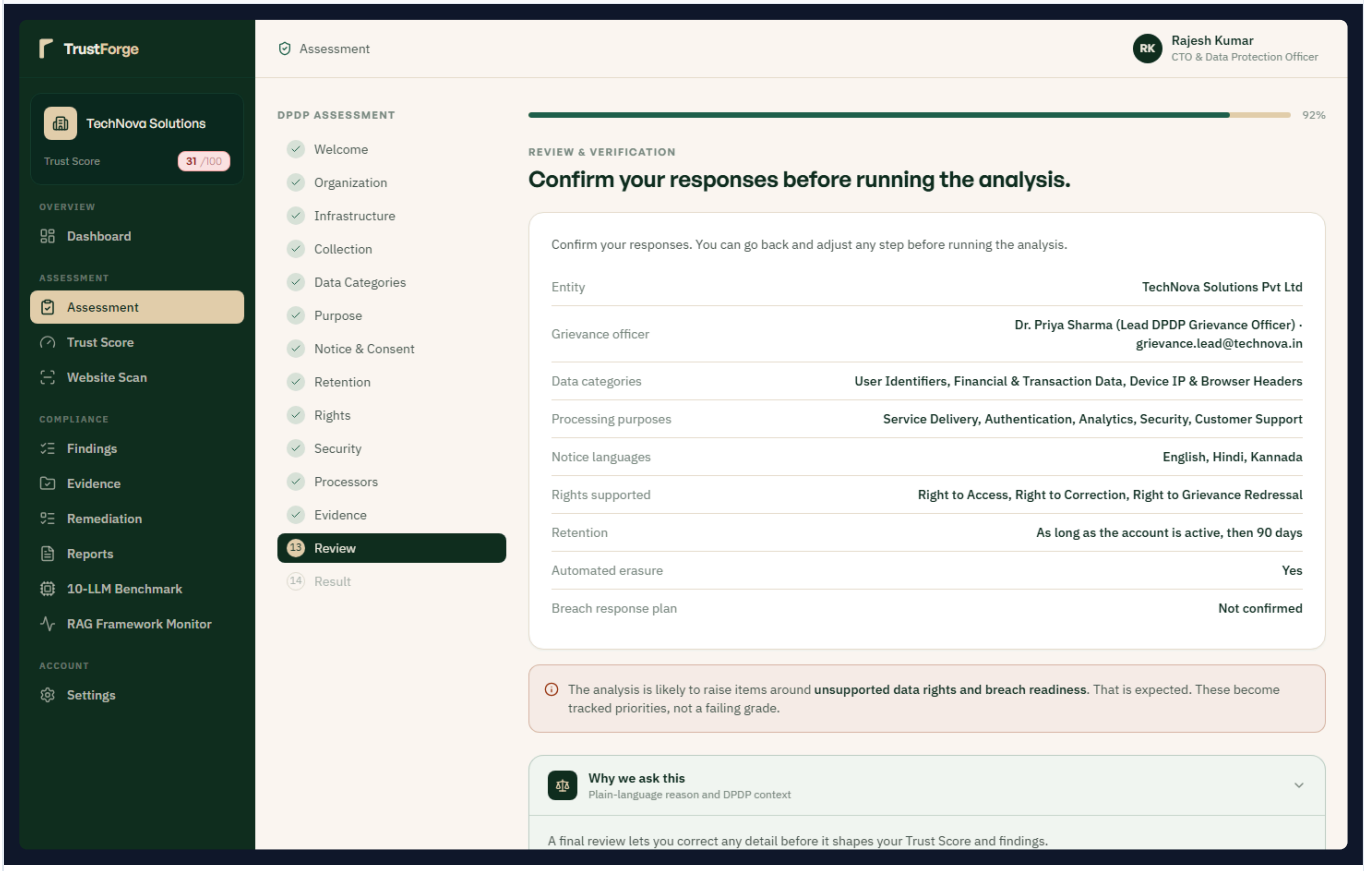
`Route: /assessment/wizard/step-12` Role: Compliance Assessor



Evaluates internal grievance redressal procedures and Data Protection Officer escalation channels under Section 13. Verifies ticket logging systems that record complaints and track resolution turnaround times. Checks that communications inform data principals of their right to appeal unresolved grievances to the Data Protection Board.

Figure 21: Assessment Step 13 — Significant Data Fiduciary (SDF) Obligations

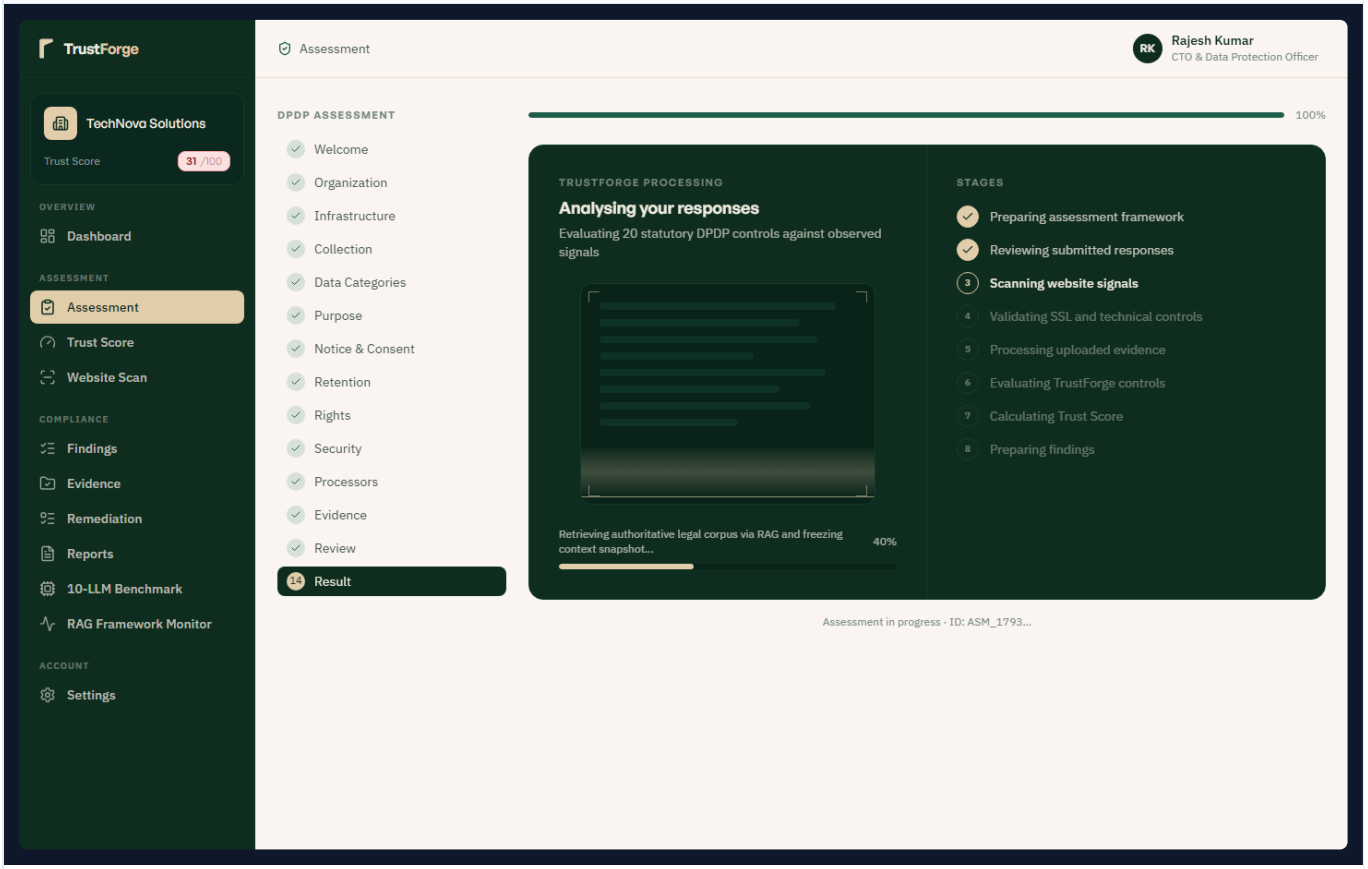
`Route: /assessment/wizard/step-13` Role: Compliance Assessor



Audits higher-tier obligations applicable to designated Significant Data Fiduciaries under DPDP Act Section 10. Assesses periodic Data Protection Impact Assessment (DPIA) frameworks, independent audit readiness, and resident DPO appointments. Reviews algorithmic risk monitoring procedures and board reporting cadences for systemic privacy impacts.

Figure 22: Assessment Step 14 — Final Audit Review & Evidence Synthesis

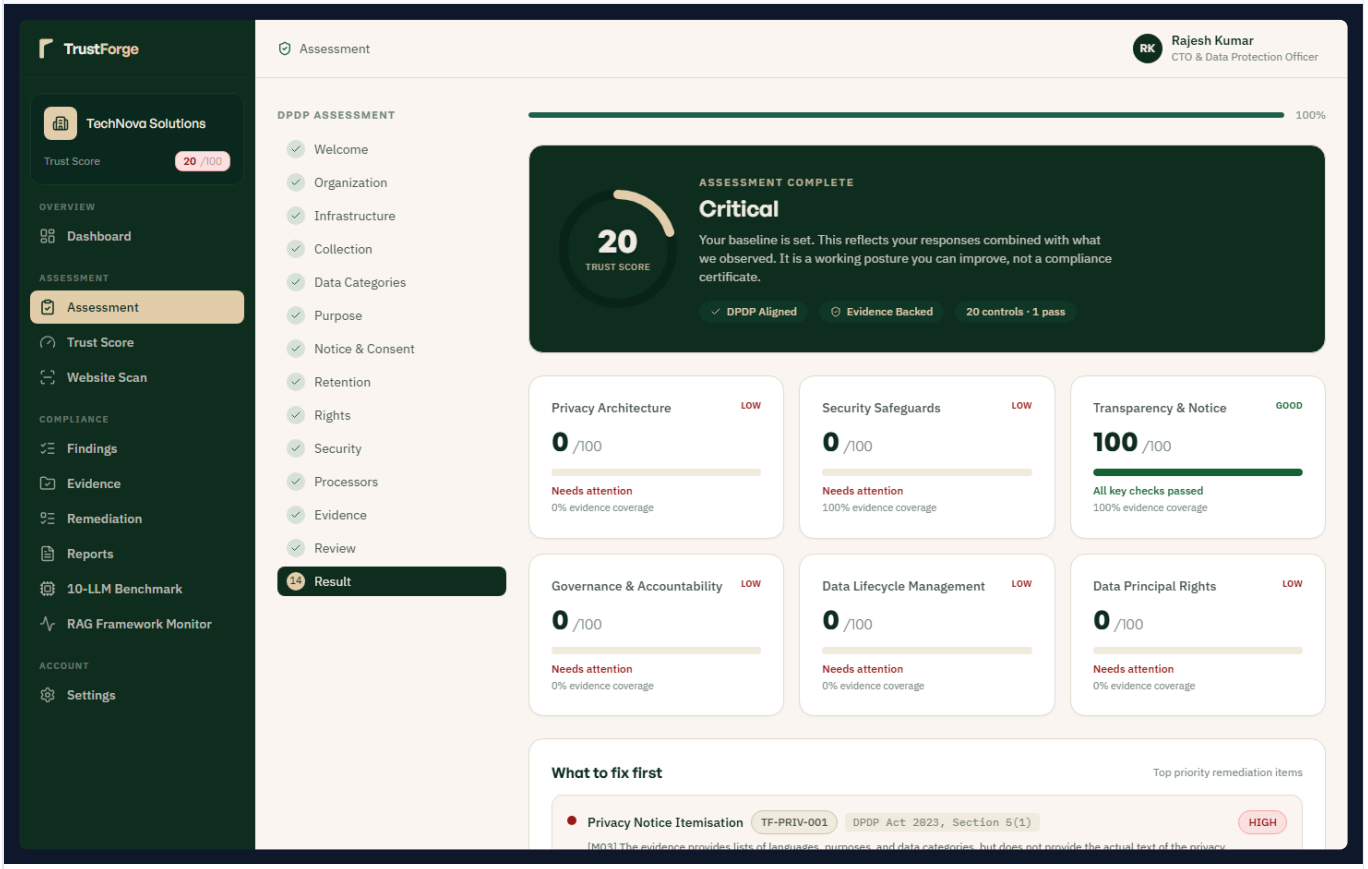
`Route: /assessment/wizard/step-14` Role: Compliance Assessor



Consolidates responses from all 14 statutory control sections into a pre-submission review matrix with completeness indicators. Allows the compliance assessor to inspect consistency, attach supplementary documentation, and confirm submission accuracy. Final submission locks the assessment version and initiates deterministic scoring and RAG citation validation.

Figure 23: Complete Statutory Compliance Evaluation Results

`Route: /assessment/results/:id` Role: Compliance Officer / DPO

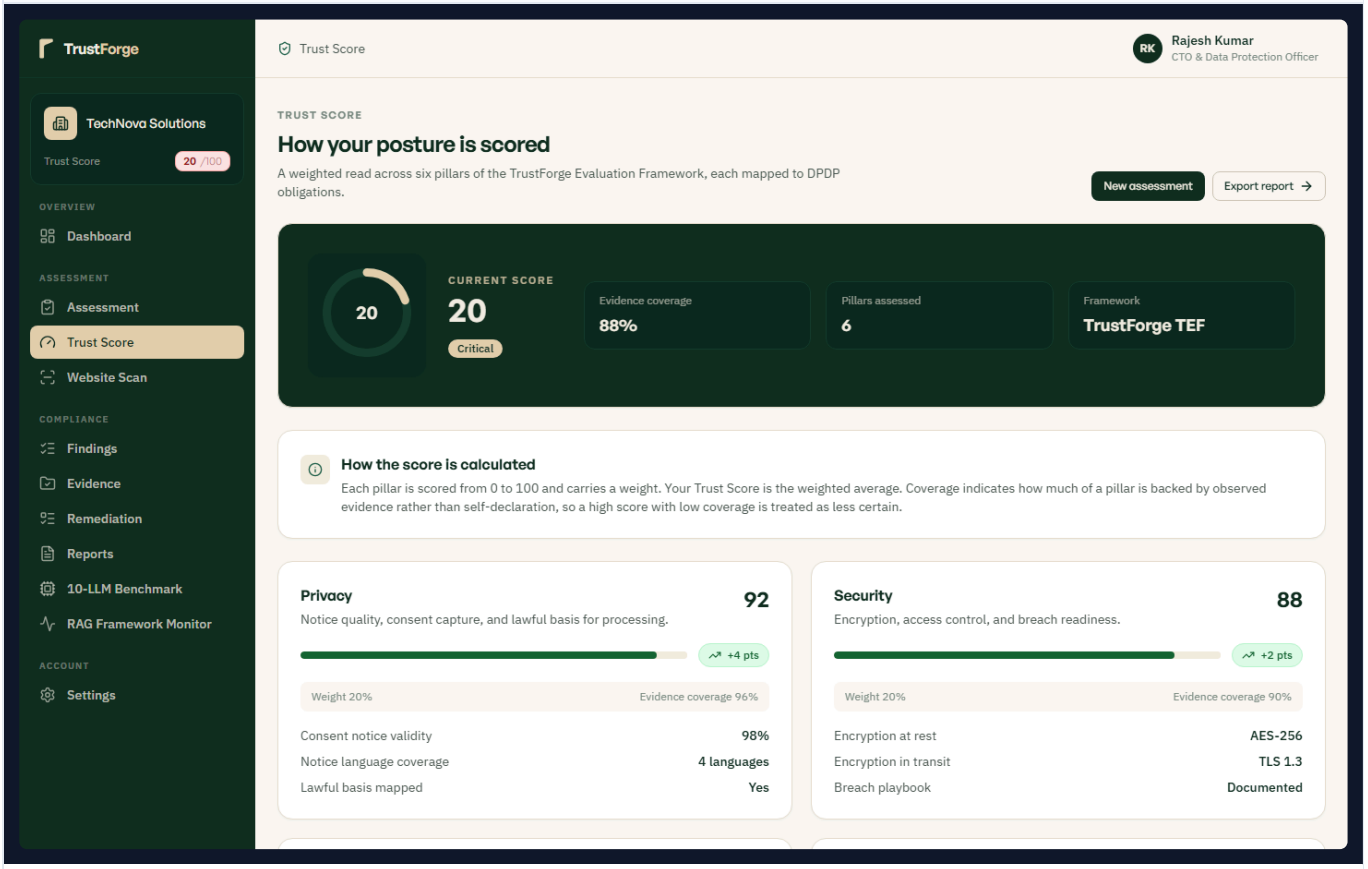


Presents the completed audit scorecard, section-by-section compliance percentages, and identified statutory gap summaries. Categorizes compliance gaps by severity (High, Medium, Low) based on statutory obligations and potential exposure tiers. Generates actionable remediation recommendations and provides access to printable summary reports.

Compliance Engines & Analytics

Figure 24: TrustForge Composite Trust Score & Empirical Analytics

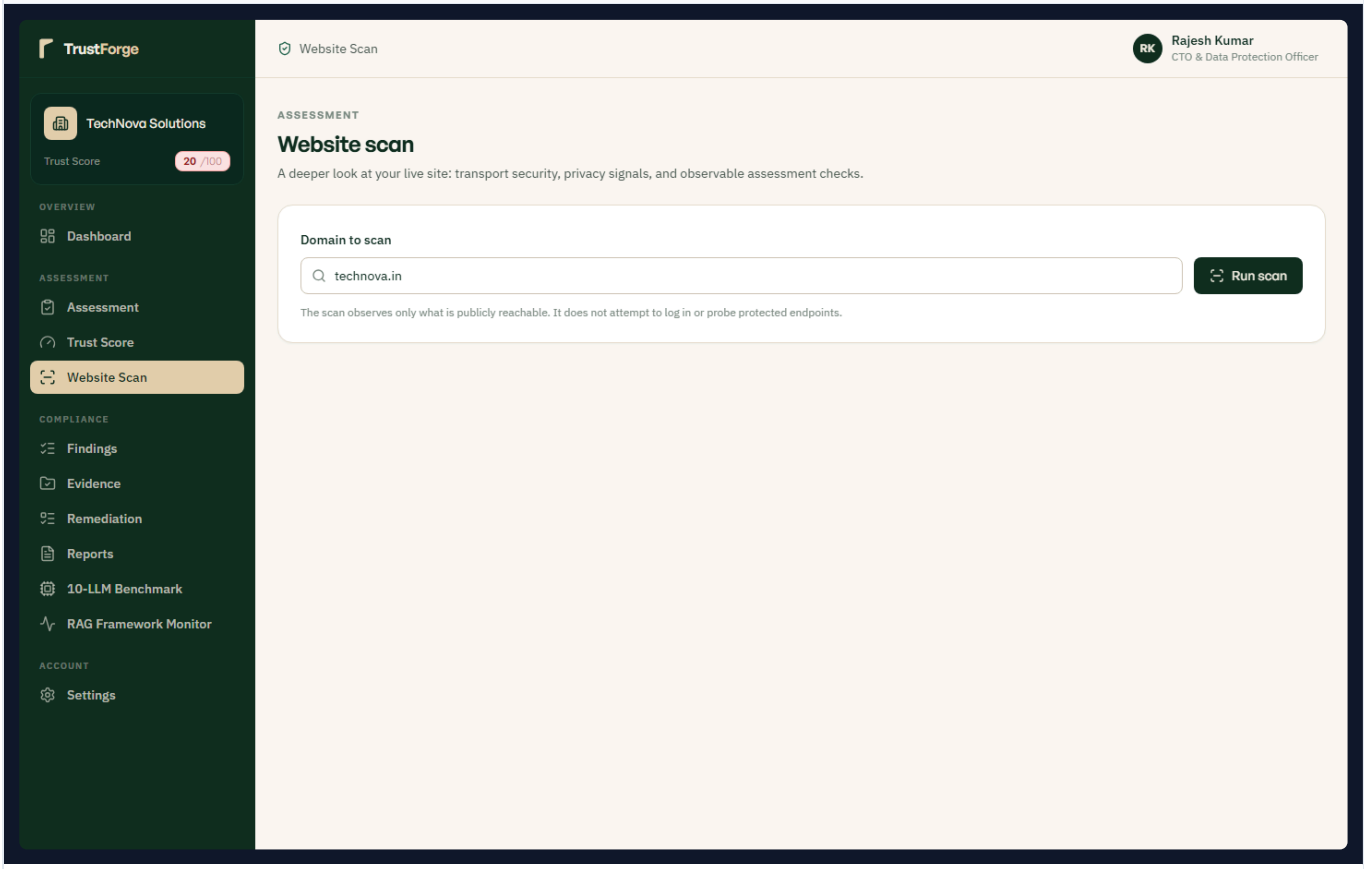
`Route: /trust-score` Role: Executive / Auditor



Displays the deterministic Trust Score computed across six statutory pillars: Privacy, Security, Transparency, Governance, Lifecycle, Rights. Incorporates faculty C1–C10 compliance mappings, dual weighting schemes (W1 Penalty-Proportional, W2 Equal), and posture bands. Enforces critical gating caps (capping scores at 40 if HTTPS or privacy notices are missing) and provides pillar radar visualizations.

Figure 25: Automated Perimeter Security & Privacy Cookie Scanner

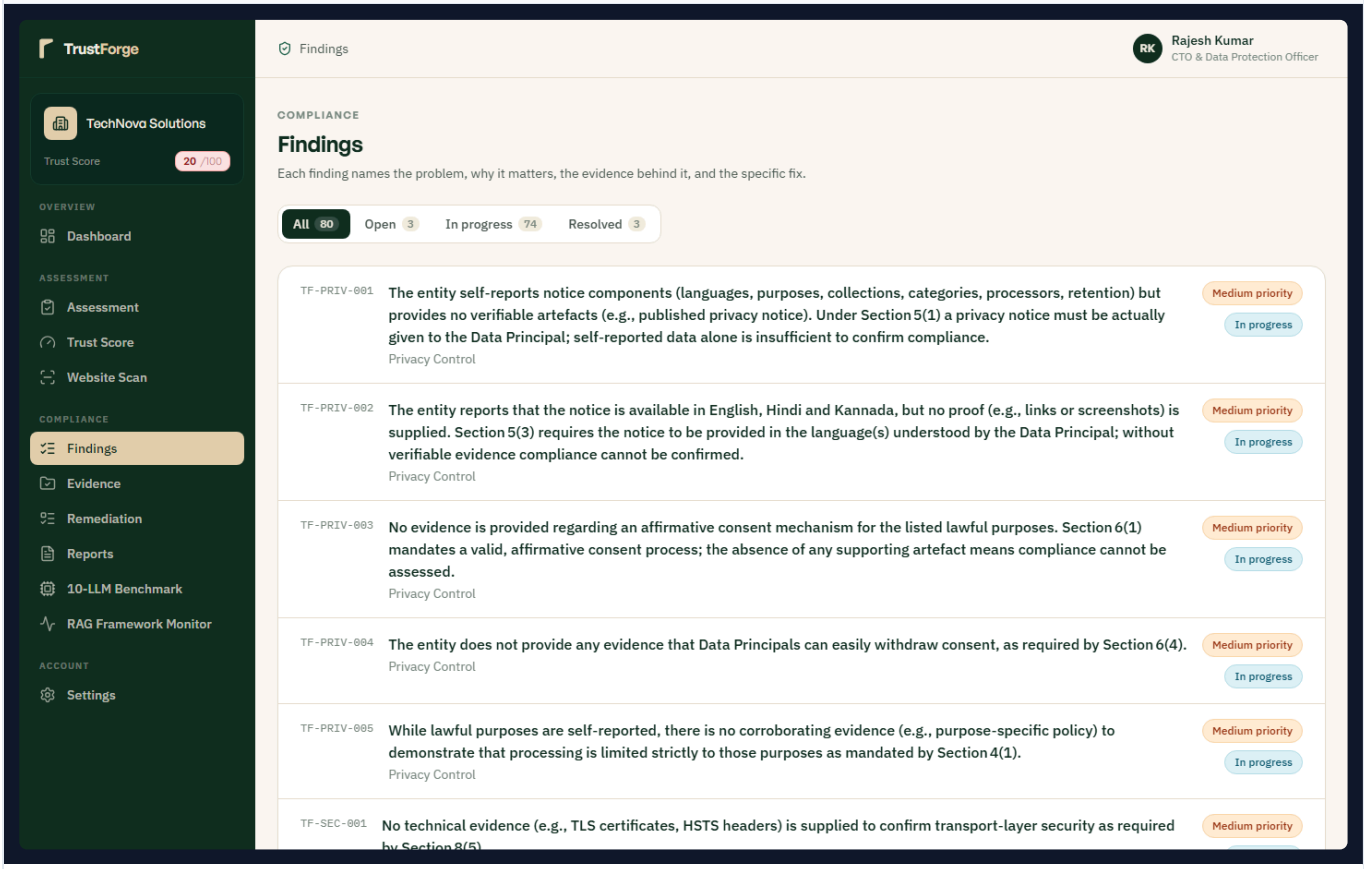
`Route: /scans` Role: Security Engineer / Auditor



Executes automated analysis of verified enterprise web properties to identify tracking cookies and missing security headers. Evaluates SSL/TLS transport availability, defensive HTTP headers (HSTS, CSP), and accessibility of privacy notices. Observed perimeter findings are automatically linked into the findings vault for remediation tracking.

Figure 26: Statutory Vulnerabilities & Findings Management Vault

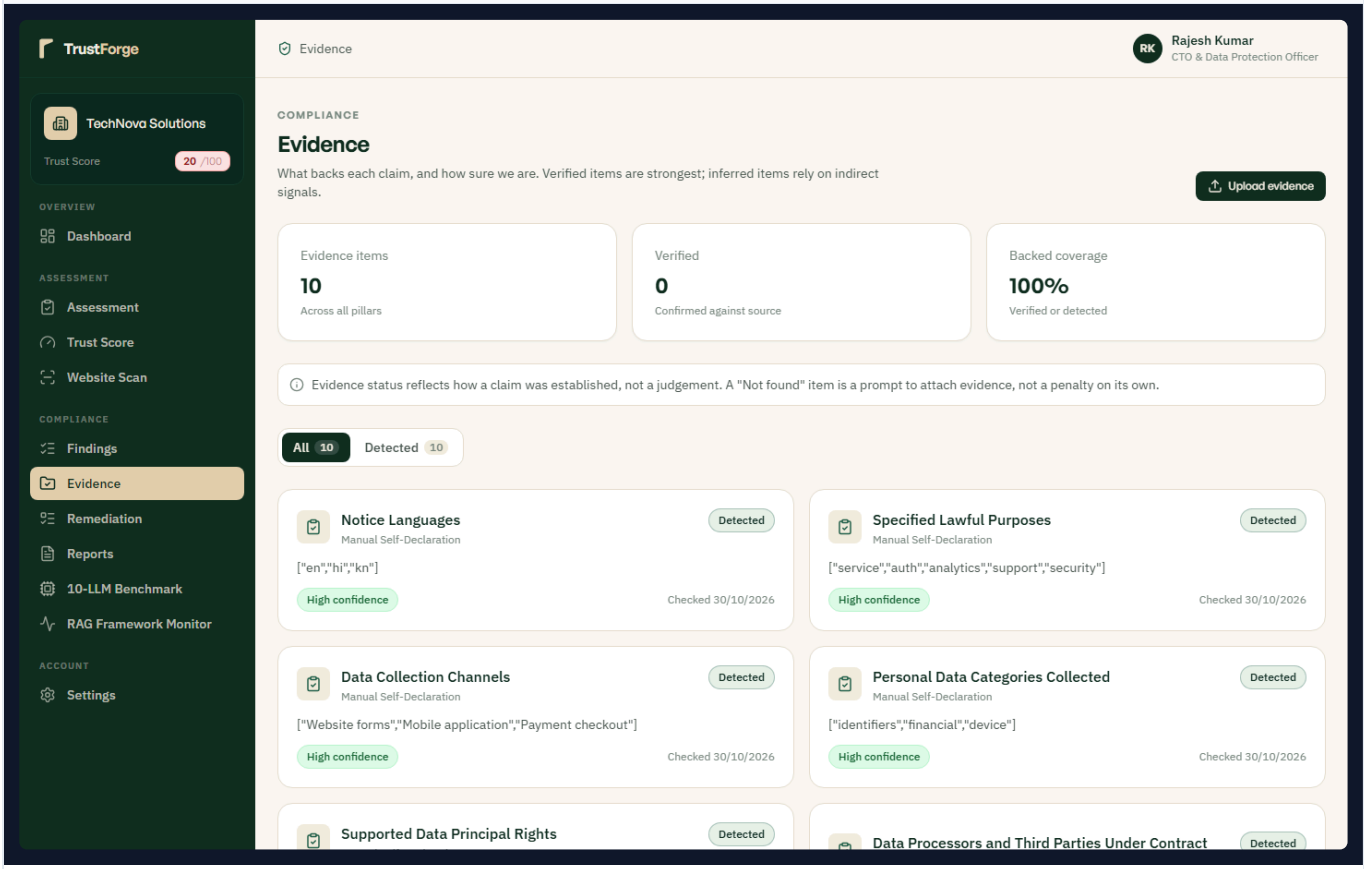
`Route: /findings` Role: Compliance Manager



Centralizes compliance gaps and perimeter scan findings in a searchable issue tracker categorized by statutory section and severity. Links each item to relevant DPDP Act provisions, descriptive remediation notes, and current resolution status. Allows compliance managers to assign remediation owners and update remediation progress.

Figure 27: Cryptographic Evidence Locker & Document Management

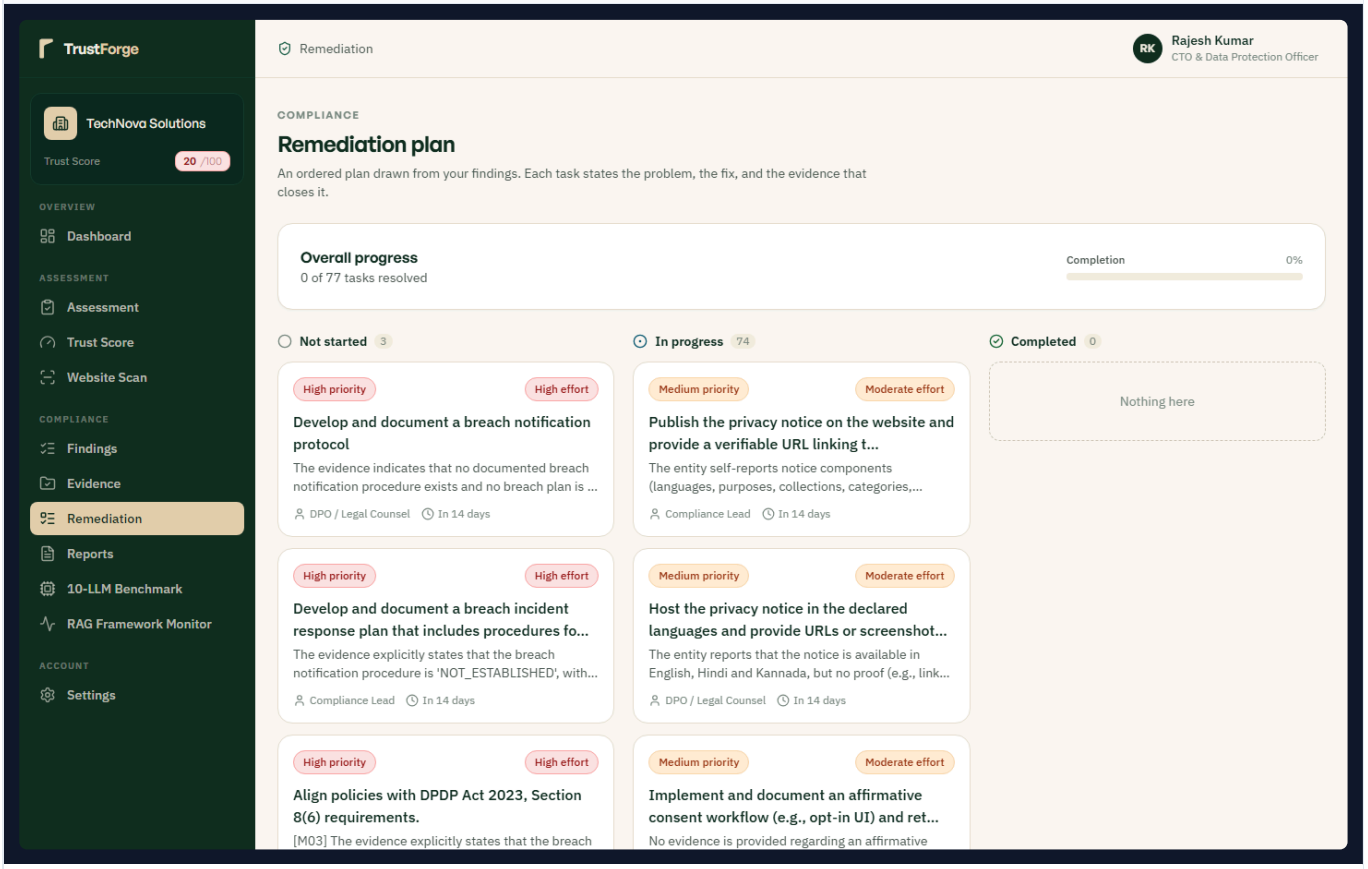
`Route: /evidence` Role: Compliance Officer



Provides an auditable evidence locker where organizations upload policy documents, consent logs, and architecture documentation. Each uploaded file is hashed using SHA-256 upon ingestion to establish a tamper-evident chain of custody and file verification record. Stored evidence is indexed for retrieval to support citation grounding during automated compliance assessments.

Figure 28: Interactive Statutory Remediation Roadmaps & Workflows

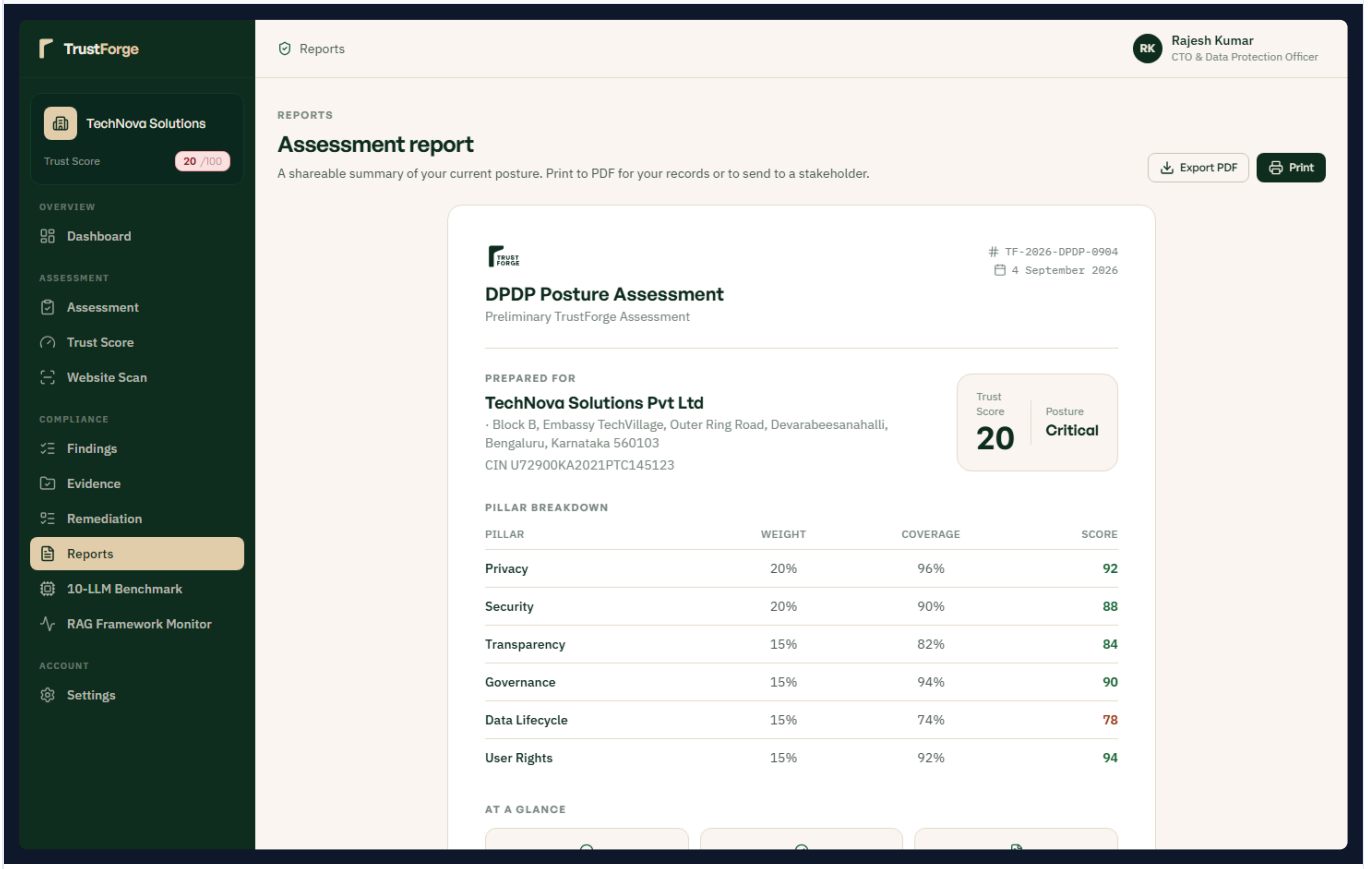
`Route: /remediation` Role: Remediation Lead / Engineer



Translates compliance findings into an interactive task-based engineering and policy remediation roadmap. Tracks implementation progress across technical controls, policy updates, and operational procedure revisions. Provides contextual guidance and statutory references to assist internal teams in closing identified gaps.

Figure 29: Statutory Audit Report Generation & Export Console

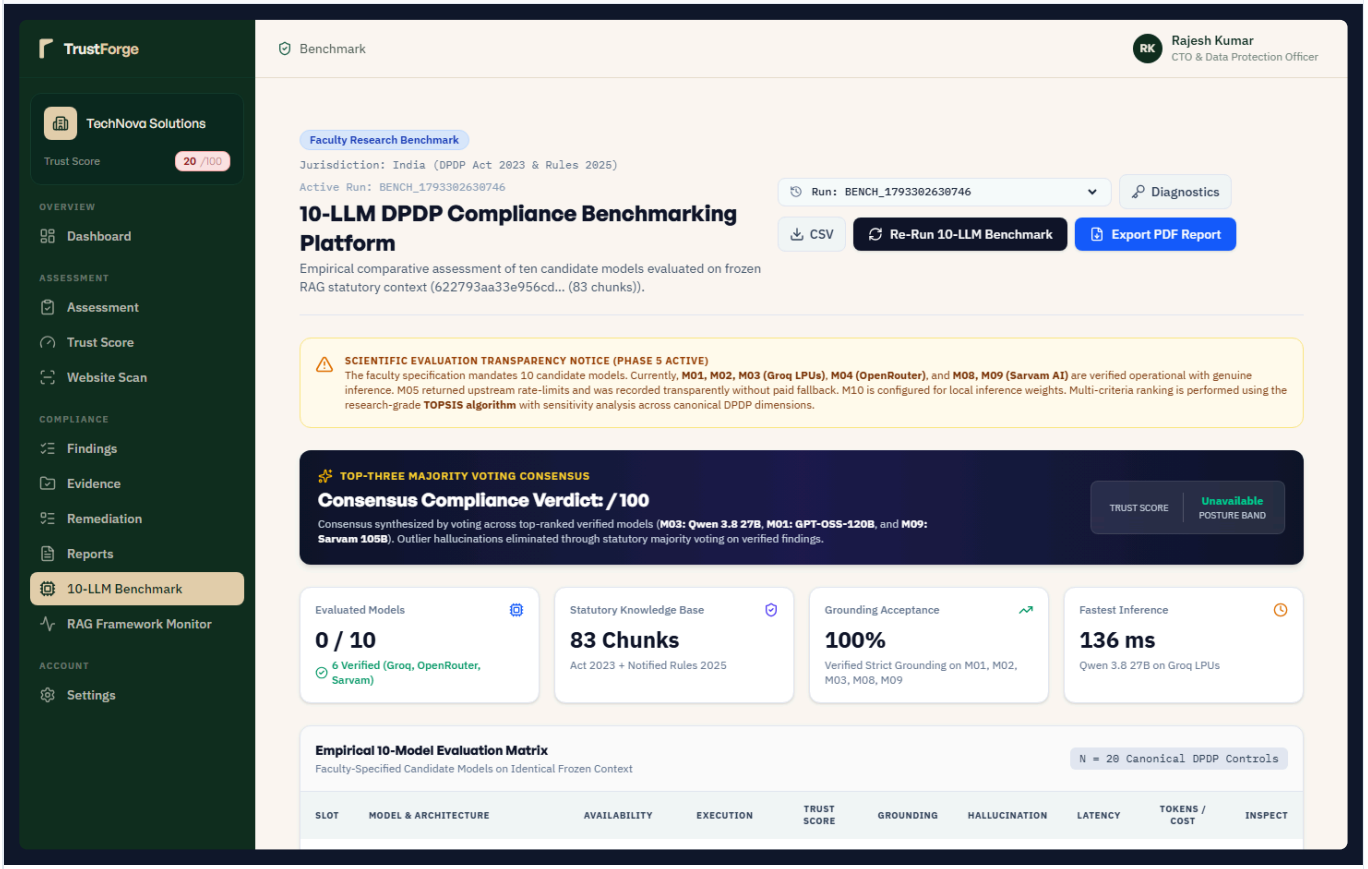
`Route: /reports` Role: Auditor / Executive



Enables export of internal compliance assessment reports in PDF and structured JSON data formats for governance review. Includes executive summaries, pillar score breakdowns, statutory section citations, and open finding inventories. Allows filtering reports by assessment date and scope for internal stakeholder review and audit preparation.

Figure 30: Multi-Model Evaluation & TOPSIS Decision Matrix

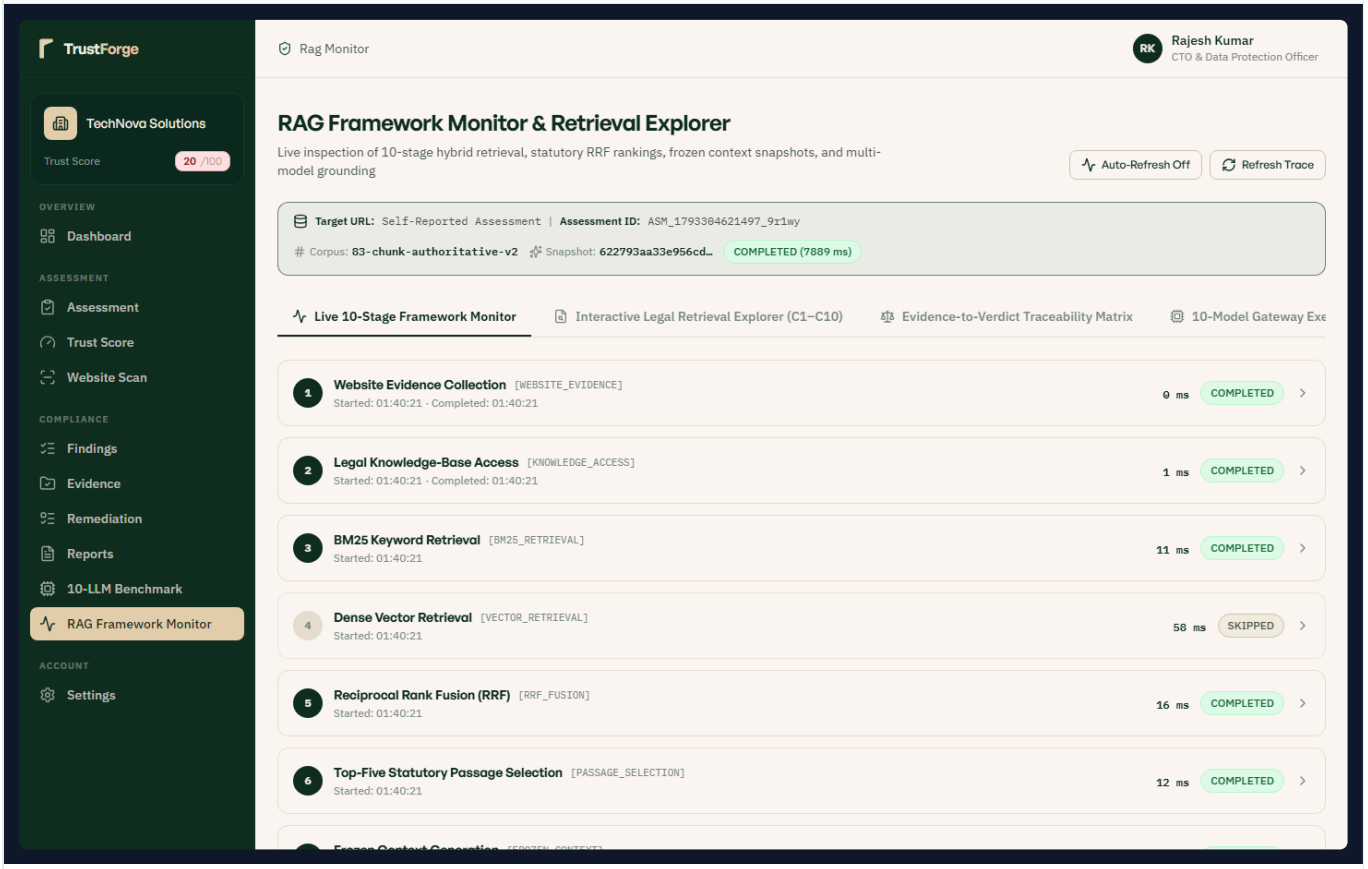
`Route: /benchmark` Role: AI Research / DPO



Presents empirical benchmark metrics across 10 designated candidate model slots (M01–M10) evaluated against DPDP Act controls. Applies the TOPSIS multi-criteria decision algorithm using 7 primary criteria (Macro-F1, Citations, Evidence, Consistency, Latency, Cost, Feasibility). Maintains zero-paid commercial API spend ($0.00) by utilizing local models and free tiers, gating billable providers M06–M09.

Figure 31: Real-Time RAG Diagnostic & Retrieval Fusion Monitor

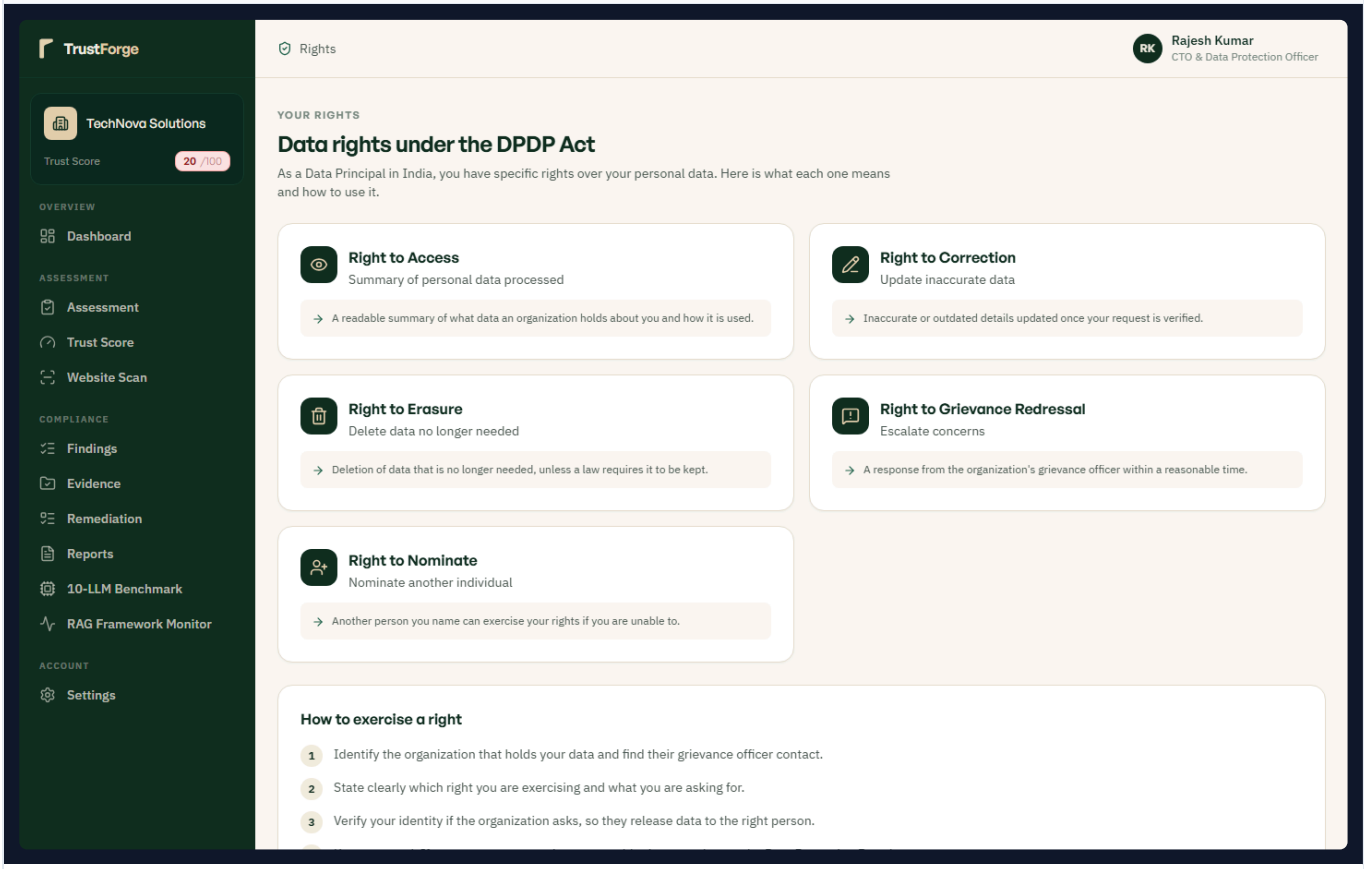
`Route: /rag-monitor` Role: Data Engineer / Auditor



Provides real-time visibility into the hybrid dense/sparse retrieval pipeline combining vector search with lexical BM25. Monitors Reciprocal Rank Fusion (RRF, k=60), cosine similarity thresholds, and statutory corpus chunk boundaries. Enables engineers to verify that model findings are grounded in retrieved excerpts from the DPDP Act 2023 legal corpus.

Figure 32: Data Principal Rights Request Management Portal

`Route: /rights-portal` Role: Privacy Operations Lead

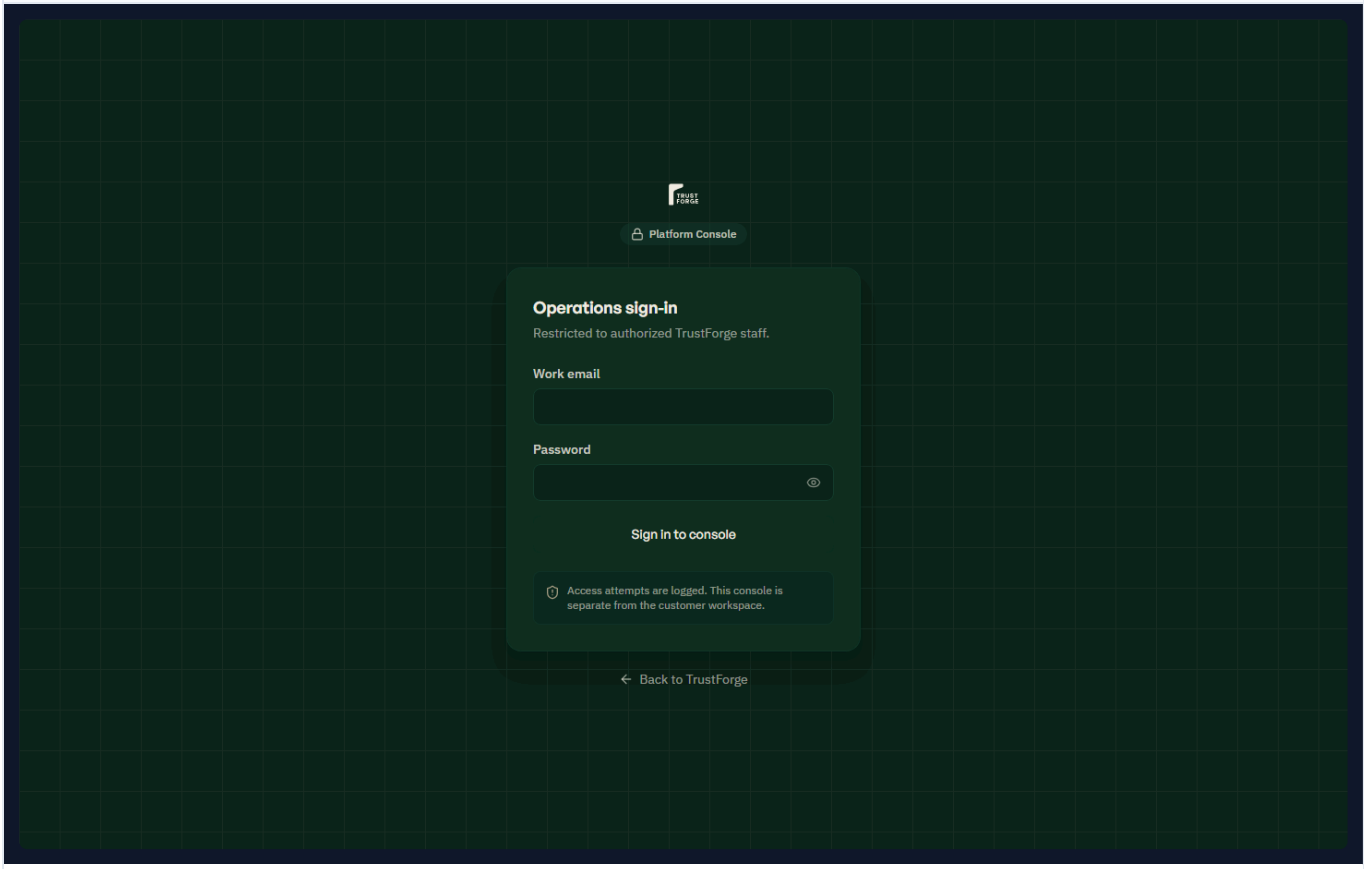


Manages incoming data principal privacy requests including Access, Correction, Erasure, and Grievance submissions. Tracks operational turnaround times against statutory 30-day response targets with visual status badges. Records request processing milestones and documentation to maintain a complete internal fulfillment history.

Administrative Governance & Platform Audit

Figure 33: Dedicated Administrative Authentication Gate

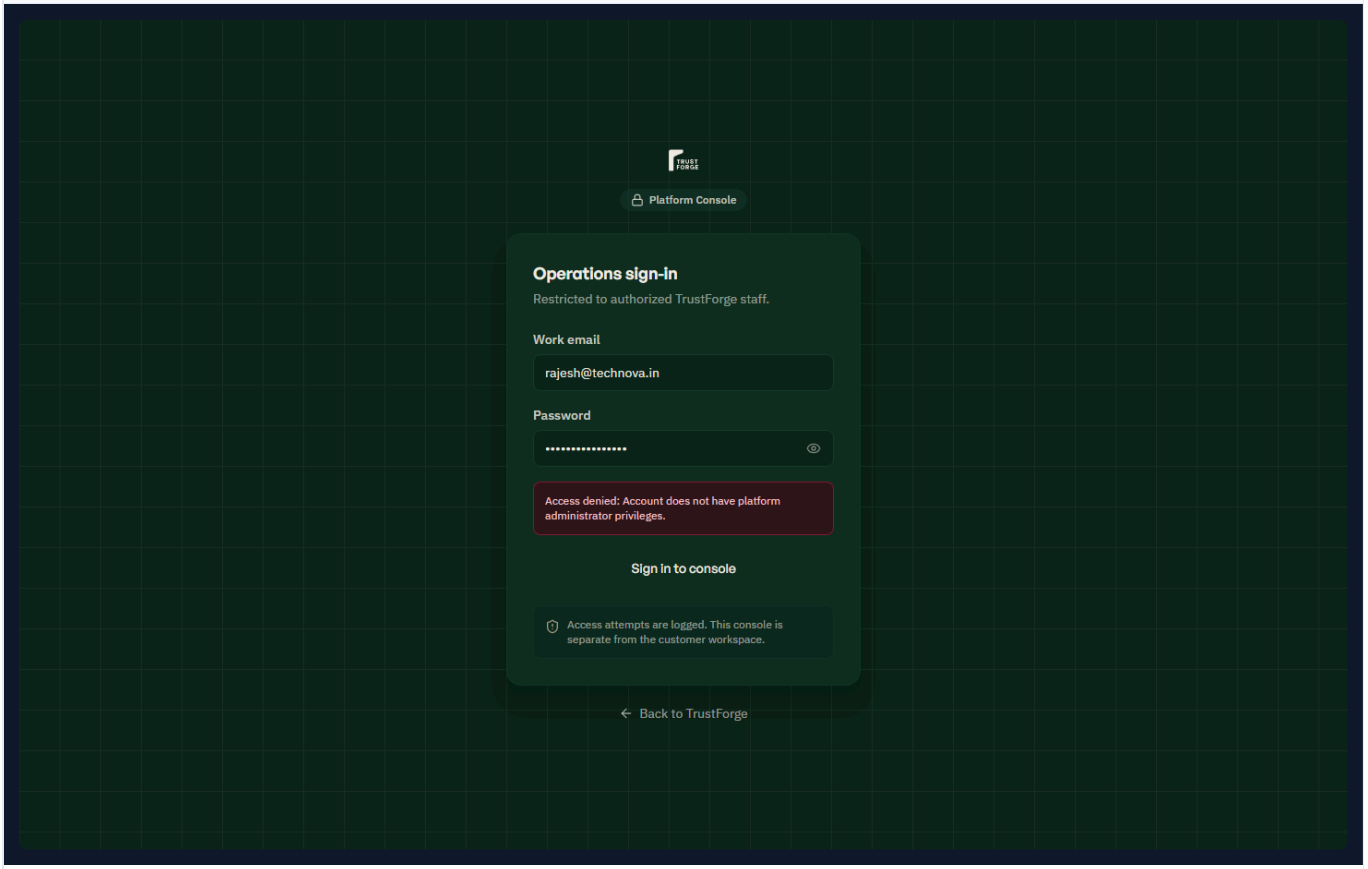
`Route: /admin/login (Dedicated)` Role: System Administrator



Enforces an isolated administrative authentication gate requiring explicit server-verified platform administrator roles. Client customer credentials submitted here are rejected by the backend to prevent unauthorized console access. Requires a non-empty password and verifies the user profile dynamically against database authority before granting access.

Figure 34: Administrative Access Control & Privilege Rejection Error

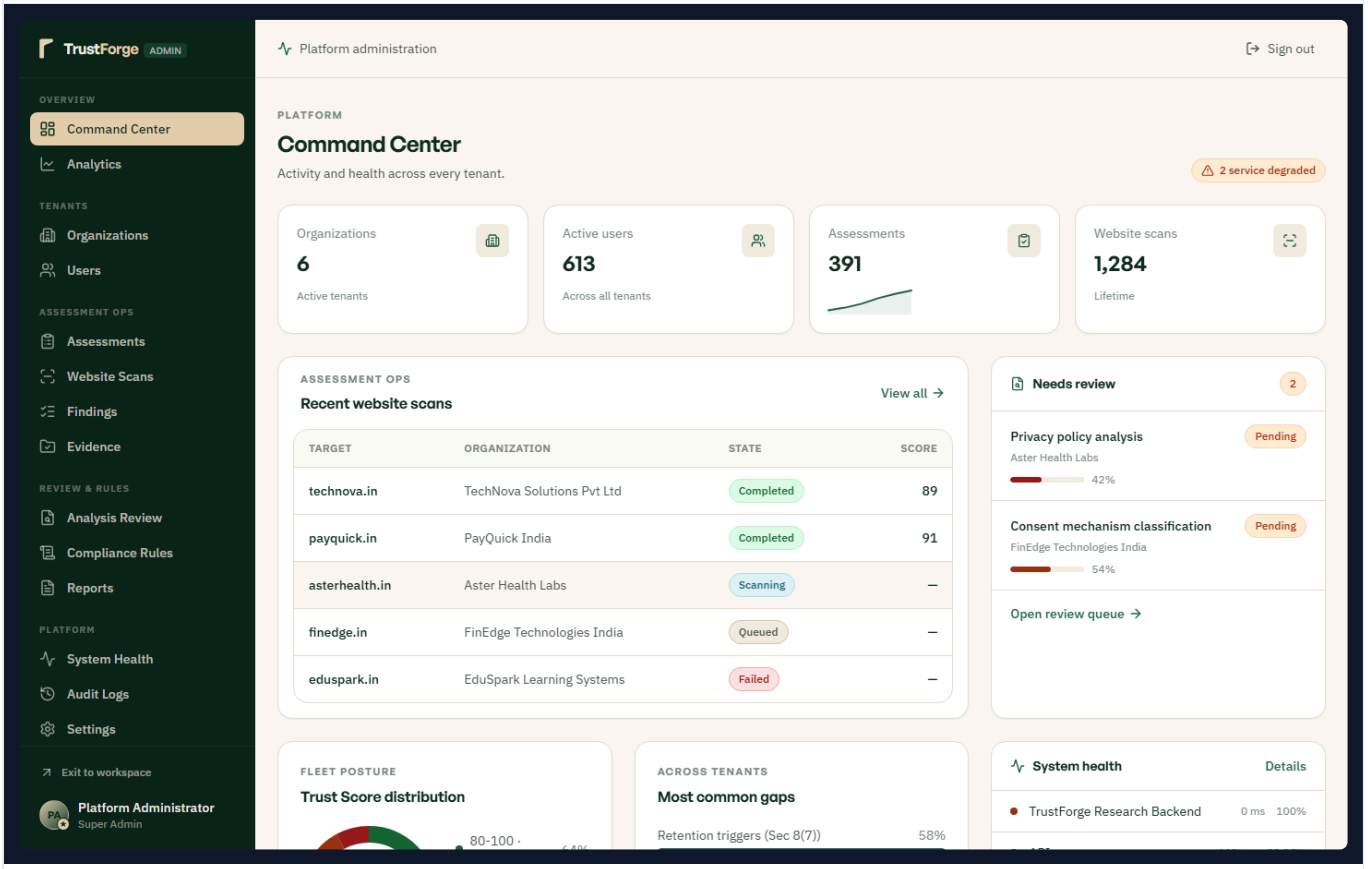
`Route: /admin/login (Failure State)` Role: Unauthorized User



Demonstrates the active access control rejection boundary when invalid or non-administrative credentials attempt console login. Displays a clear, non-leaking authentication error message while recording the unauthorized attempt in the security audit log. Prevents privilege escalation and enforces strict separation between client tenant roles and platform administrators.

Figure 35: Central Platform Command Center & Global KPIs

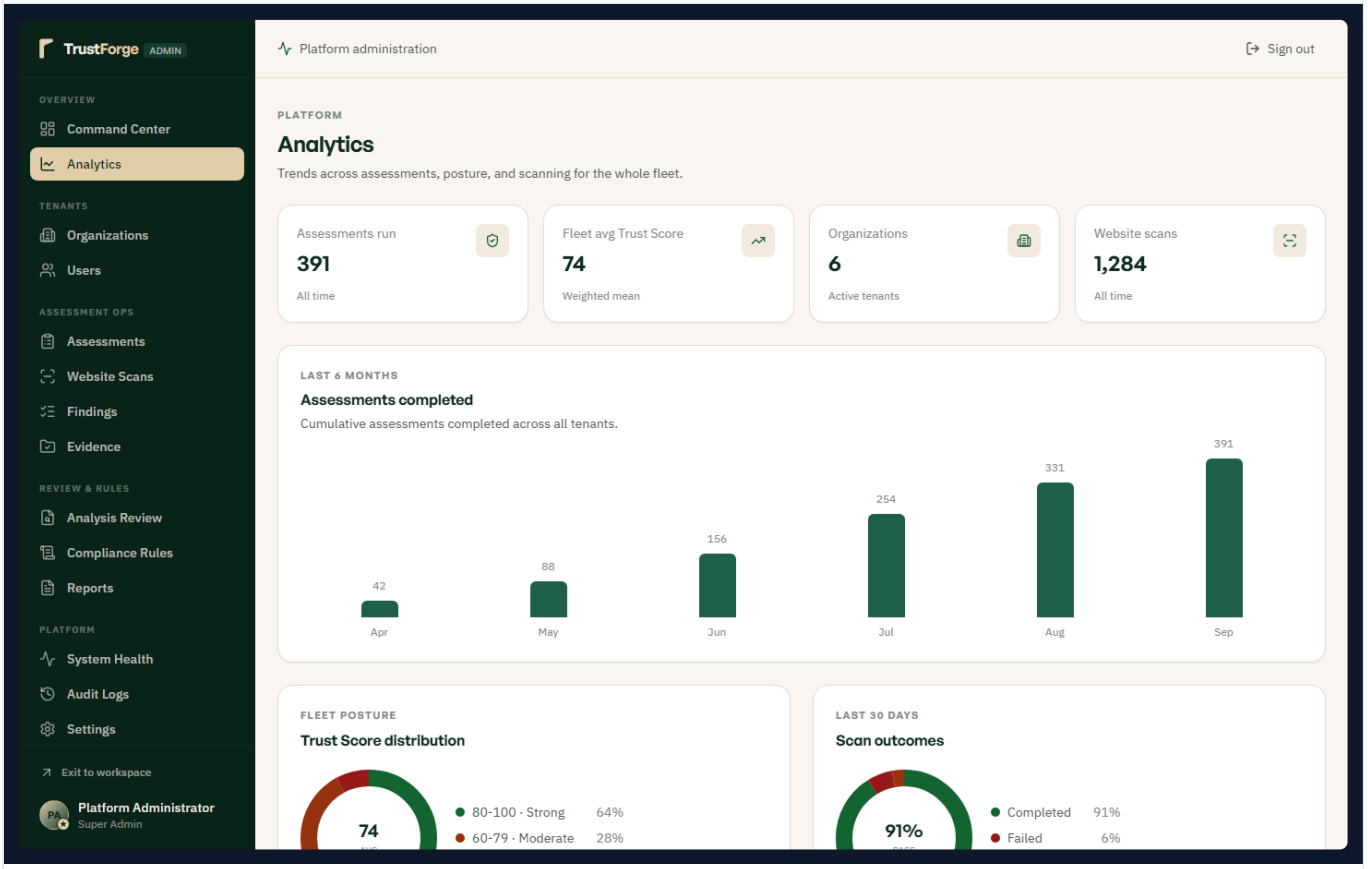
`Route: /admin/dashboard` Role: Platform Superadmin



Aggregates platform-wide telemetry, cross-tenant compliance indices, active scan queues, and aggregate assessment totals. Provides platform administrators with operational situational awareness regarding system activity and workload distribution. Displays overview charts of system throughput, scan processing times, and aggregate finding categories.

Figure 36: Cross-Tenant Compliance Analytics & Distribution Trends

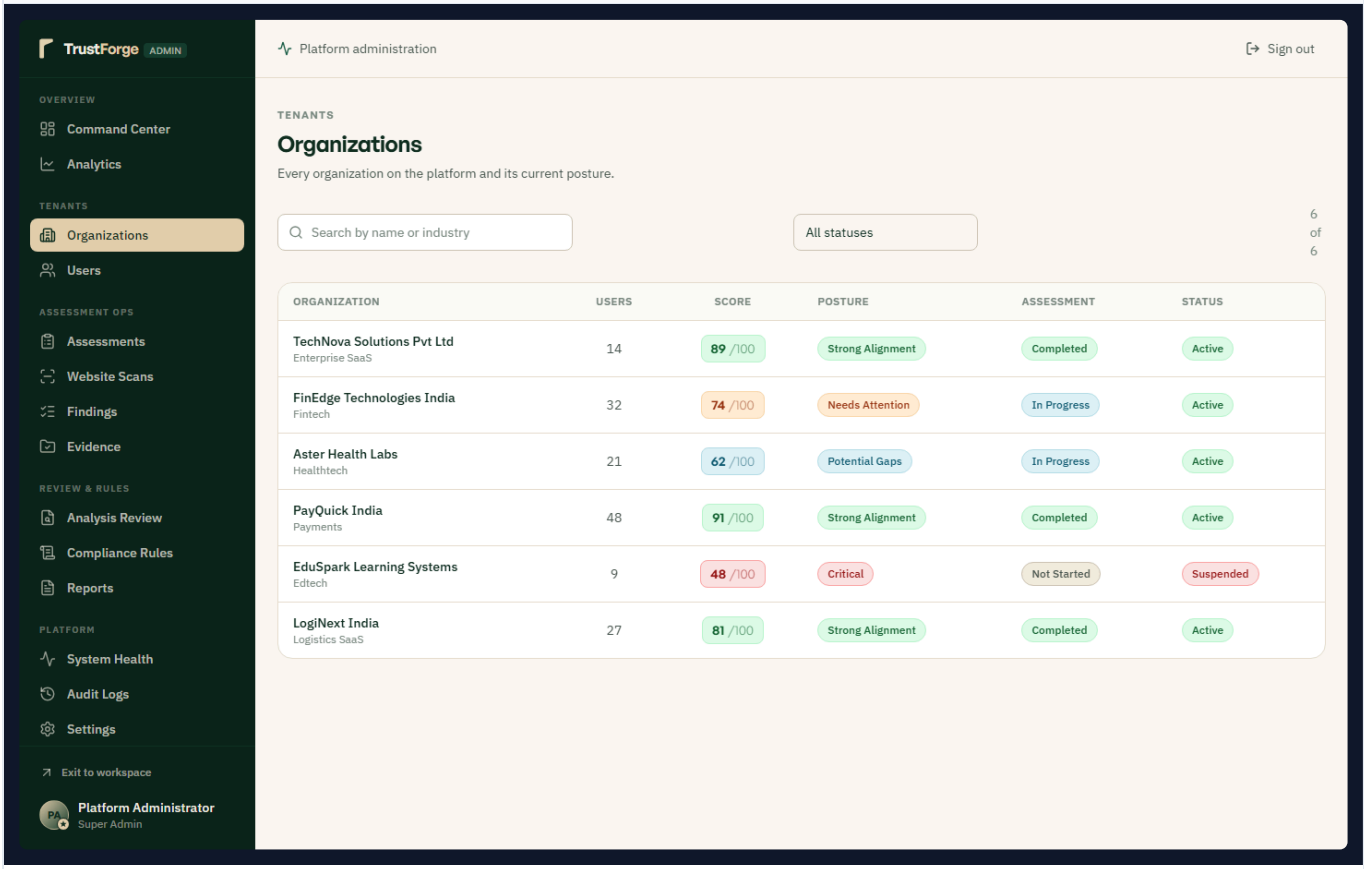
`Route: /admin/analytics` Role: Compliance Analyst



Presents aggregated cross-organization compliance trends, common statutory gap areas, and cross-sector comparisons. Visualizes the distribution of Trust Scores across industry categories (Healthcare, Finance, SaaS, Retail). Enables platform analysts to identify recurring compliance bottlenecks under the DPDP Act across registered tenants.

Figure 37: Multi-Tenant Enterprise Organization Management

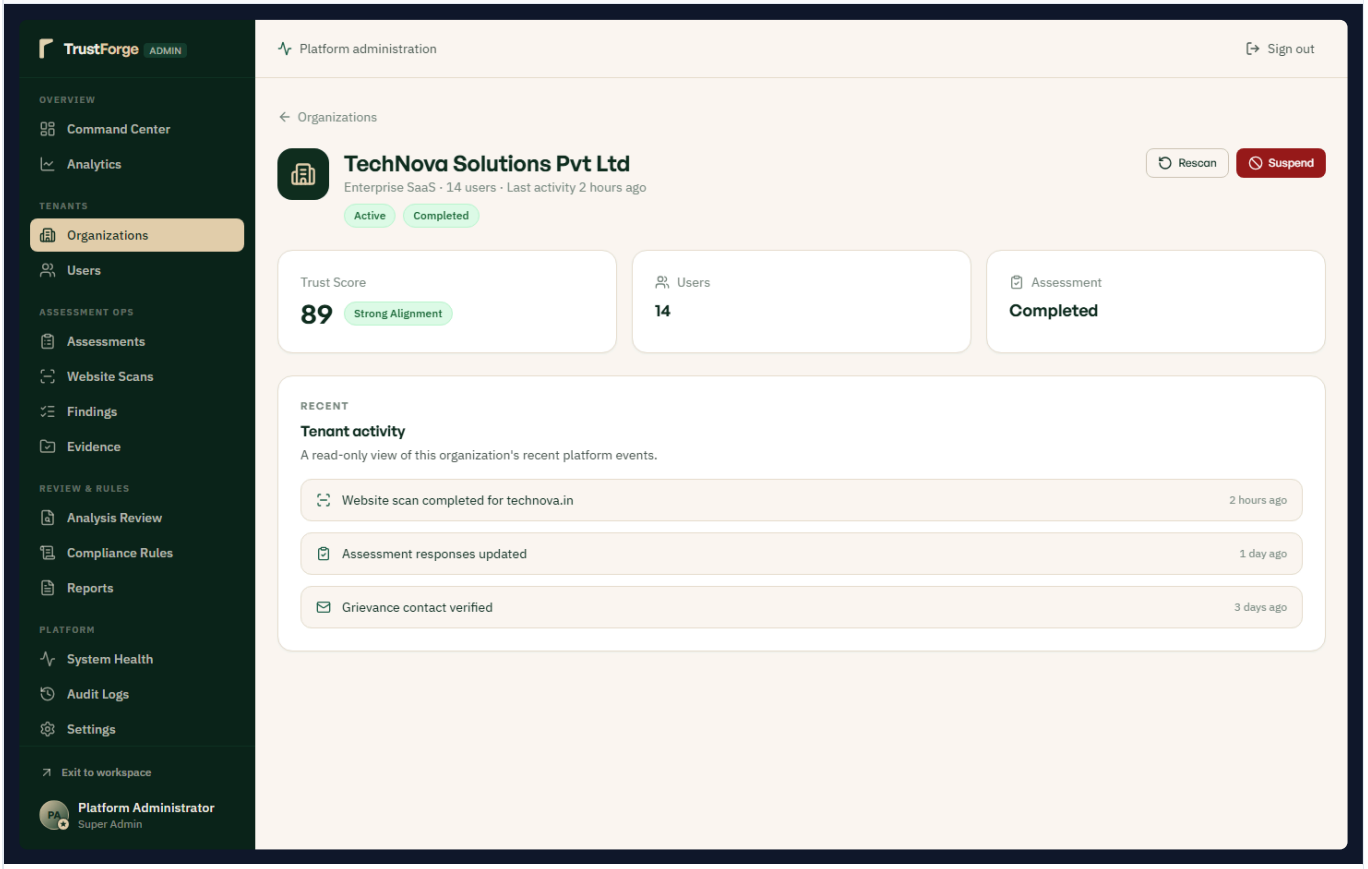
`Route: /admin/organizations` Role: Platform Superadmin



Manages registered tenant organizations, subscription tier settings, domain verification statuses, and compliance postures. Provides administrative controls to update tenant status, configure resource allocations, and inspect organization profiles. Supports searching and filtering across registered legal entities, verification flags, and Significant Data Fiduciary designations.

Figure 38: Granular Organization Tenant Governance & Deep Inspection

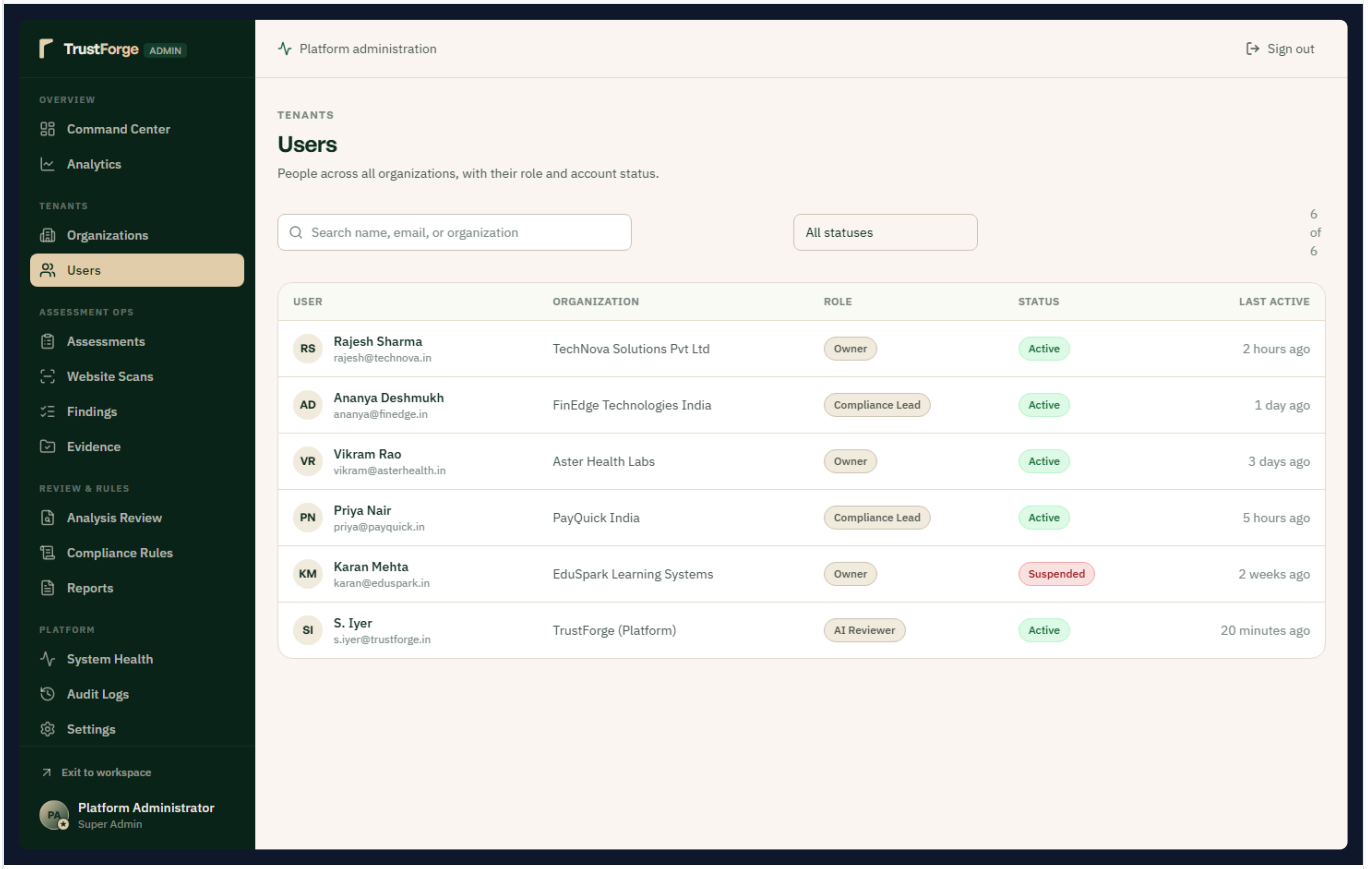
`Route: /admin/organizations/:id` Role: Platform Superadmin



Provides detailed inspection of a specific tenant's settings, assigned DPO details, DNS challenge records, and audit history. Enables manual domain verification controls for authorized enterprise testing or air-gapped organizational environments. Displays tenant-specific assessment history, registered members, and recent compliance milestones.

Figure 39: Enterprise User Directory & Role Assignment Controls

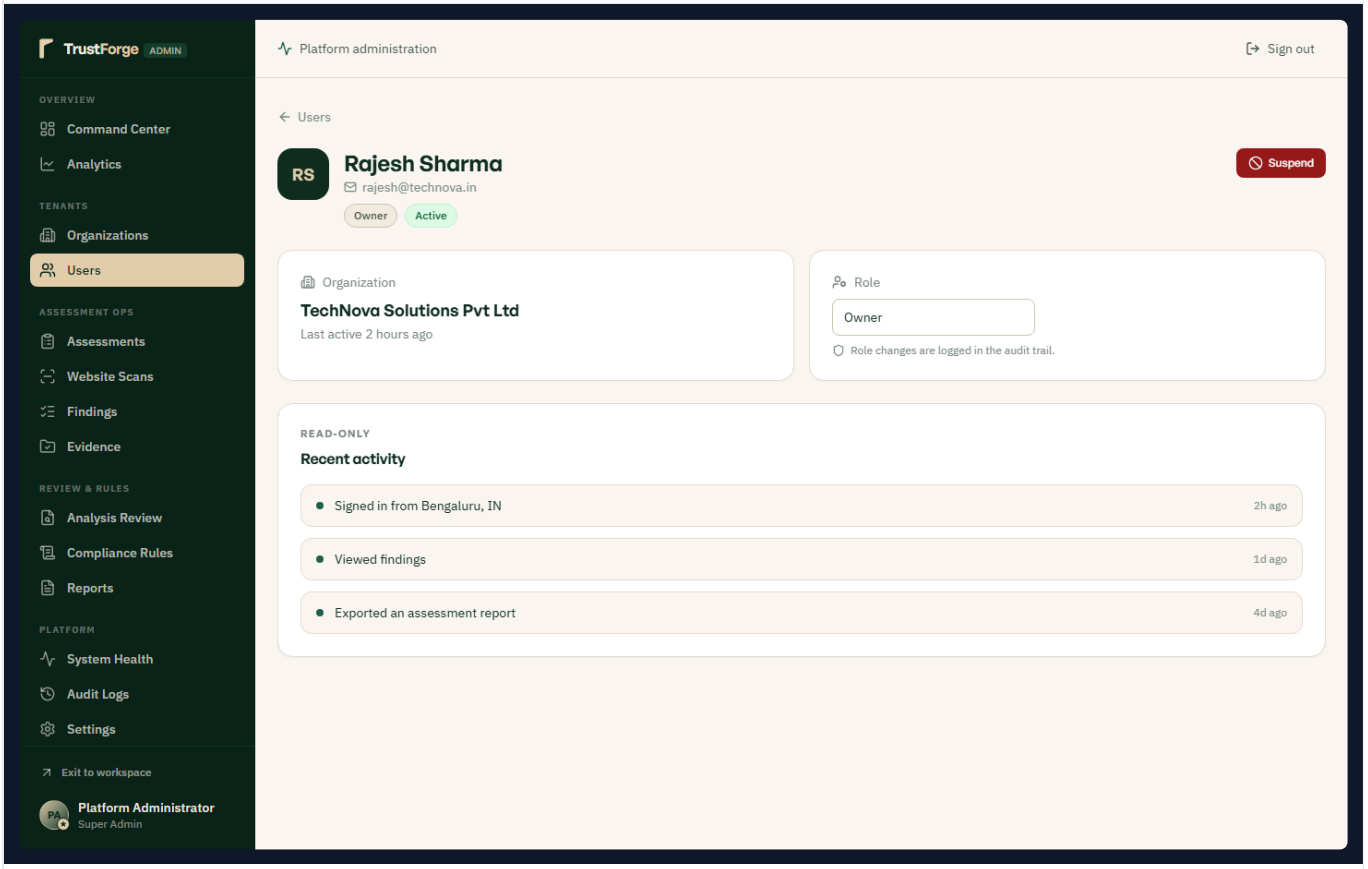
`Route: /admin/users` Role: Security Administrator



Centralizes platform user management across registered tenants, displaying assigned roles and account creation dates. Enables administrators to create user accounts, trigger password resets, and adjust role assignments (Admin, DPO, Auditor). Enforces role-based separation of responsibilities and records all administrative account changes in the audit ledger.

Figure 40: User Profile Deep Inspection & Session Revocation

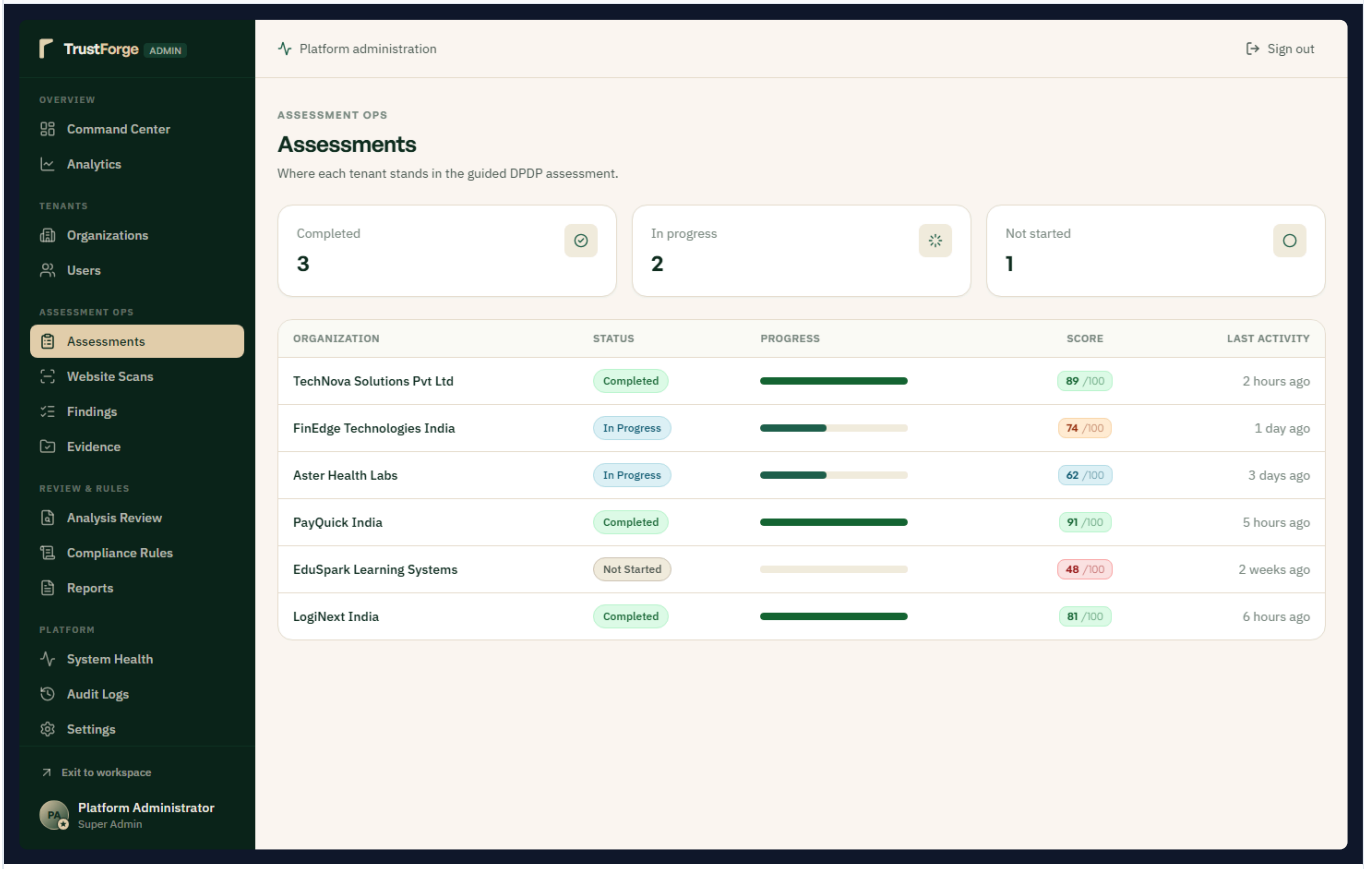
`Route: /admin/users/:id` Role: Security Administrator



Provides detailed view of individual user accounts, including associated tenant organizations, role scopes, and recent activity. Allows administrators to revoke active sessions in the event of credential rotation or employee offboarding. Audits user authentication events, account status modifications, and session lifetimes.

Figure 41: Platform-Wide Assessment Governance & Audit Logs

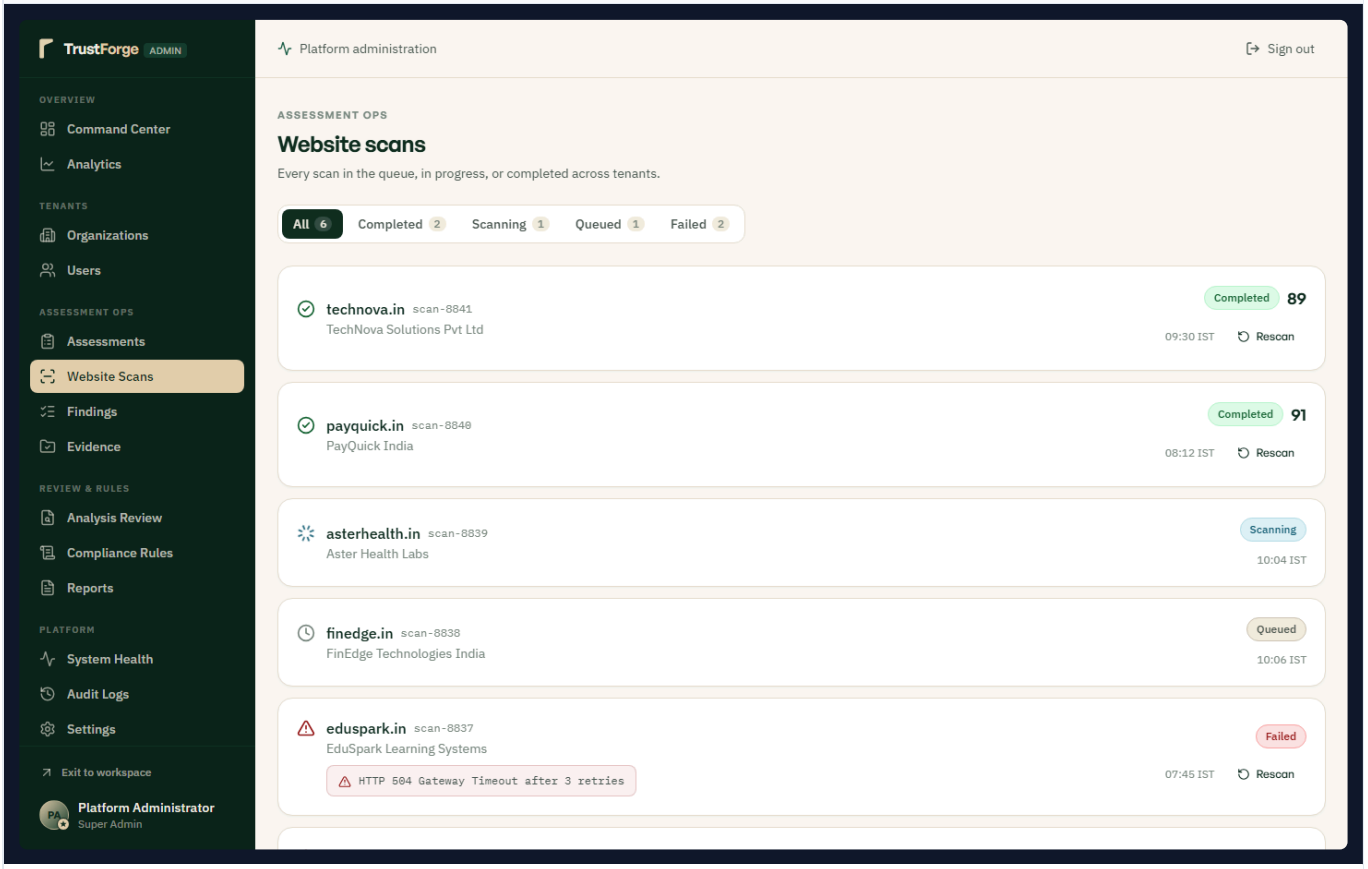
`Route: /admin/assessments` Role: Supervisory Auditor



Displays all statutory compliance assessments completed across the platform, including completion dates, scores, and review status. Enables supervisory auditors to review submitted assessment questionnaires, inspect attached evidence, and leave review notes. Allows cross-tenant compliance comparisons and drill-down into specific assessment responses.

Figure 42: Global Perimeter Scan Dispatcher & Queue Telemetry

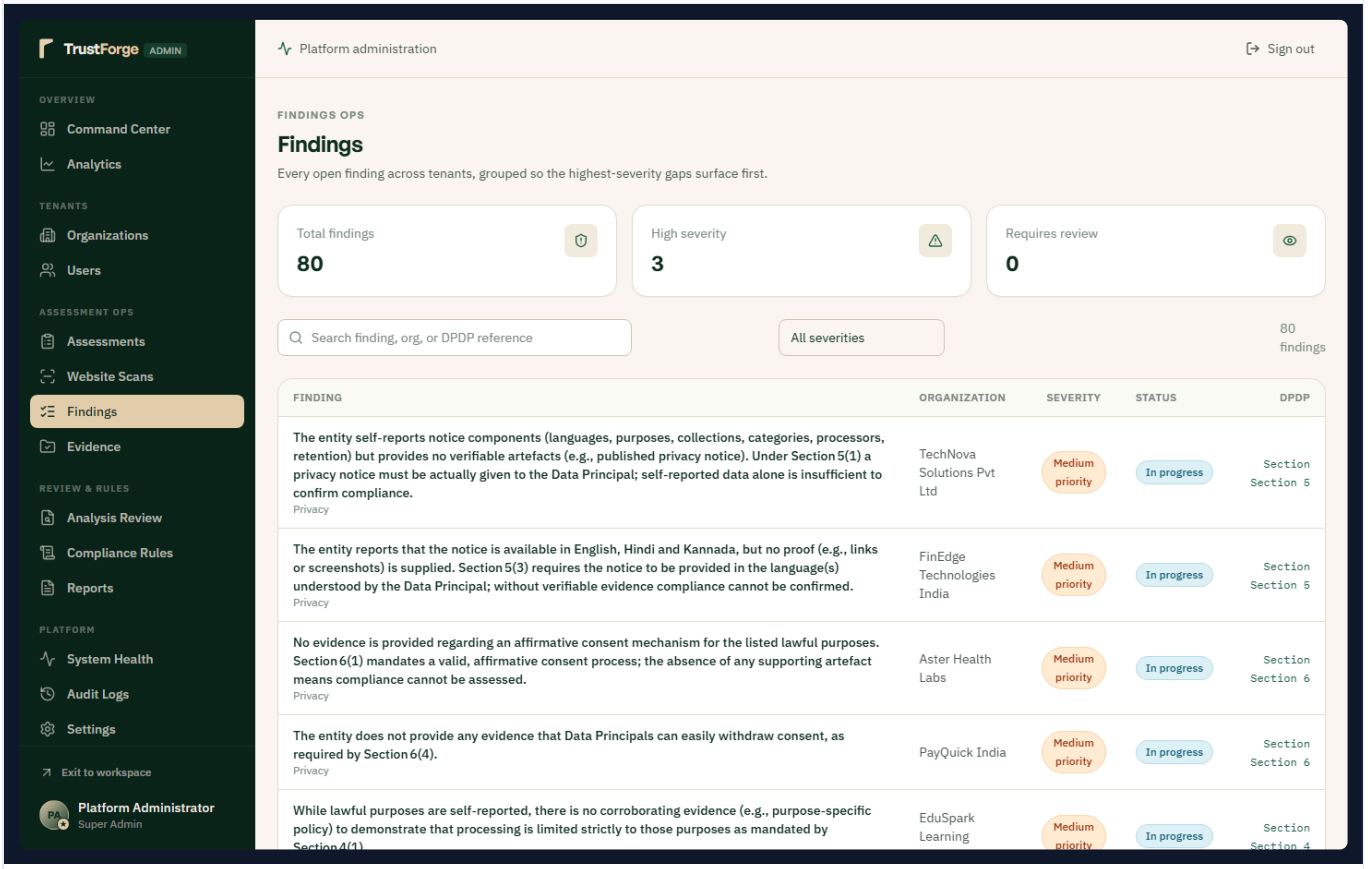
`Route: /admin/scans` Role: Systems Administrator



Monitors the background scan processing queue, showing active domain crawl tasks, completion times, and error rates. Provides administrative controls to re-run failed scan jobs and inspect scanner worker status. Logs HTTP status codes, SSL certificate parameters, and detected tracker counts for each completed scan.

Figure 43: Platform-Wide Vulnerability & Defect Registry

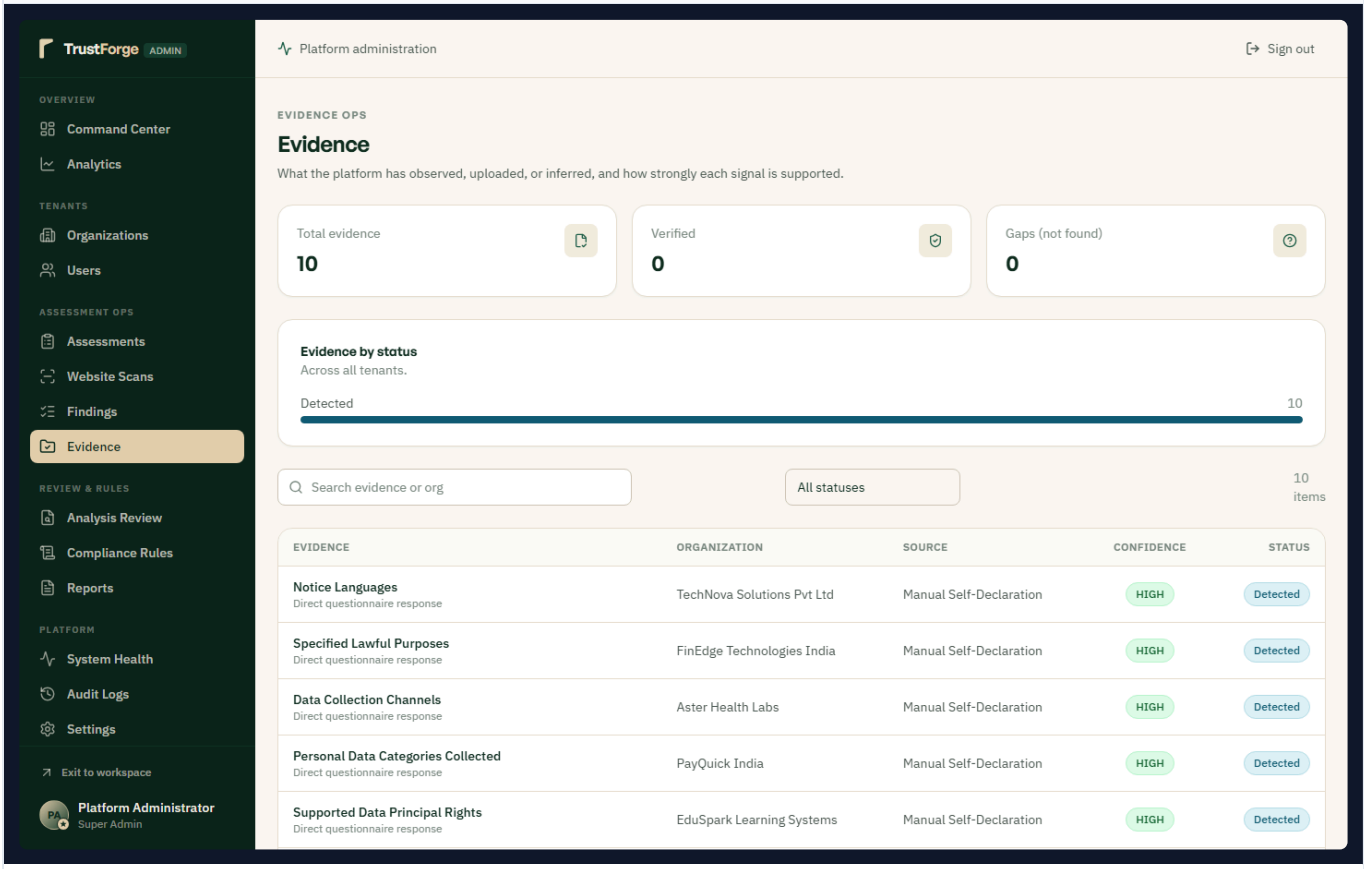
`Route: /admin/findings` Role: Chief Compliance Officer



Consolidates statutory non-compliance observations and perimeter findings across all registered organizations into one view. Allows filtering by severity level (Critical, High, Medium, Low), DPDP Act section, and current resolution status. Assists compliance leads in identifying recurring systemic issues and monitoring remediation turnaround across tenants.

Figure 44: Global Cryptographic Evidence Audit & Verification Vault

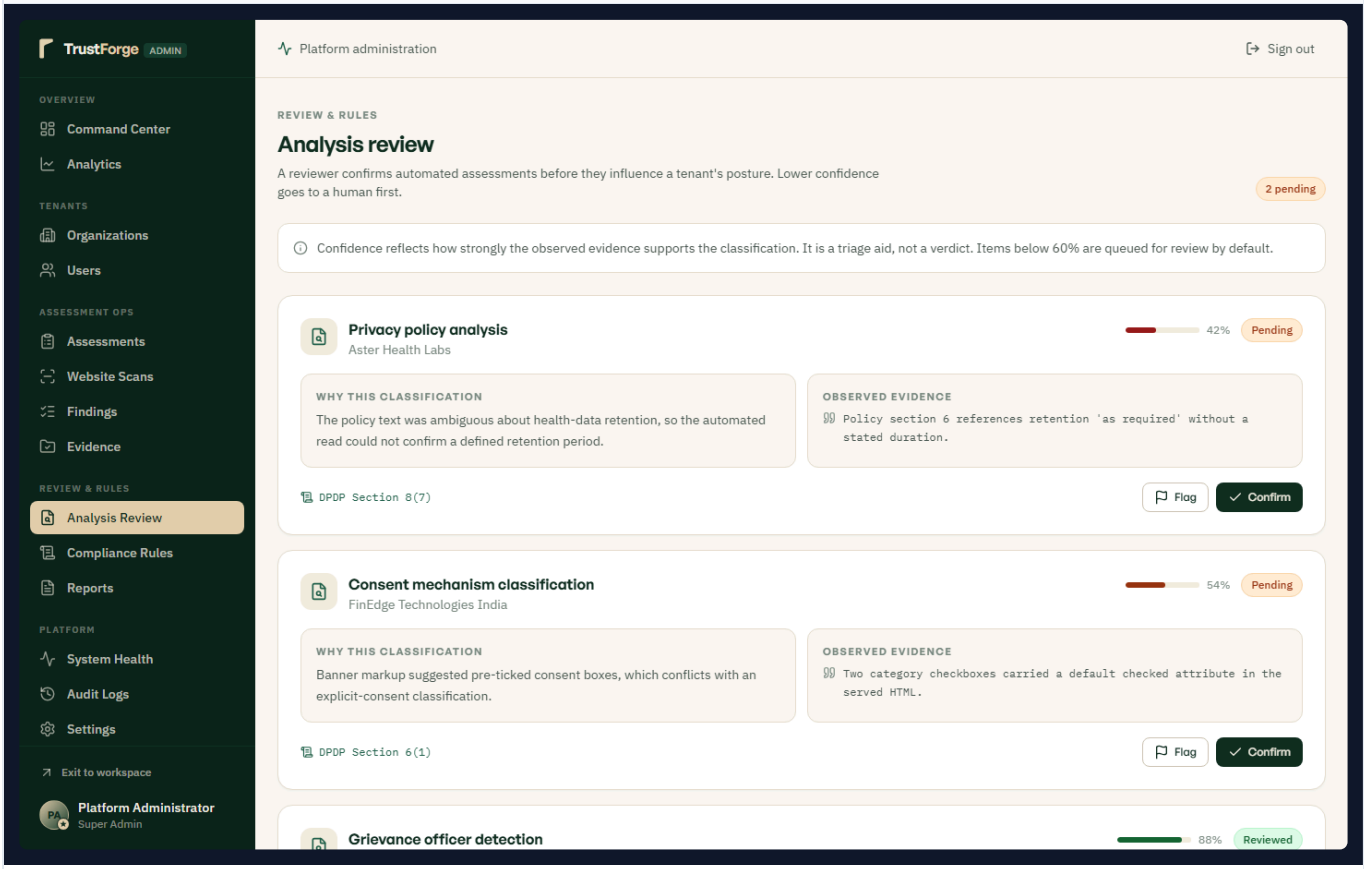
`Route: /admin/evidence` Role: Forensic Auditor



Enables platform auditors to review evidence files uploaded by tenants along with their SHA-256 cryptographic hashes. Displays file metadata, uploading tenant identity, ingestion timestamps, and linked statutory assessment sections. Verifies that uploaded documentary evidence maintains data integrity without modification post-submission.

Figure 45: Formal Compliance Review & Certification Authority

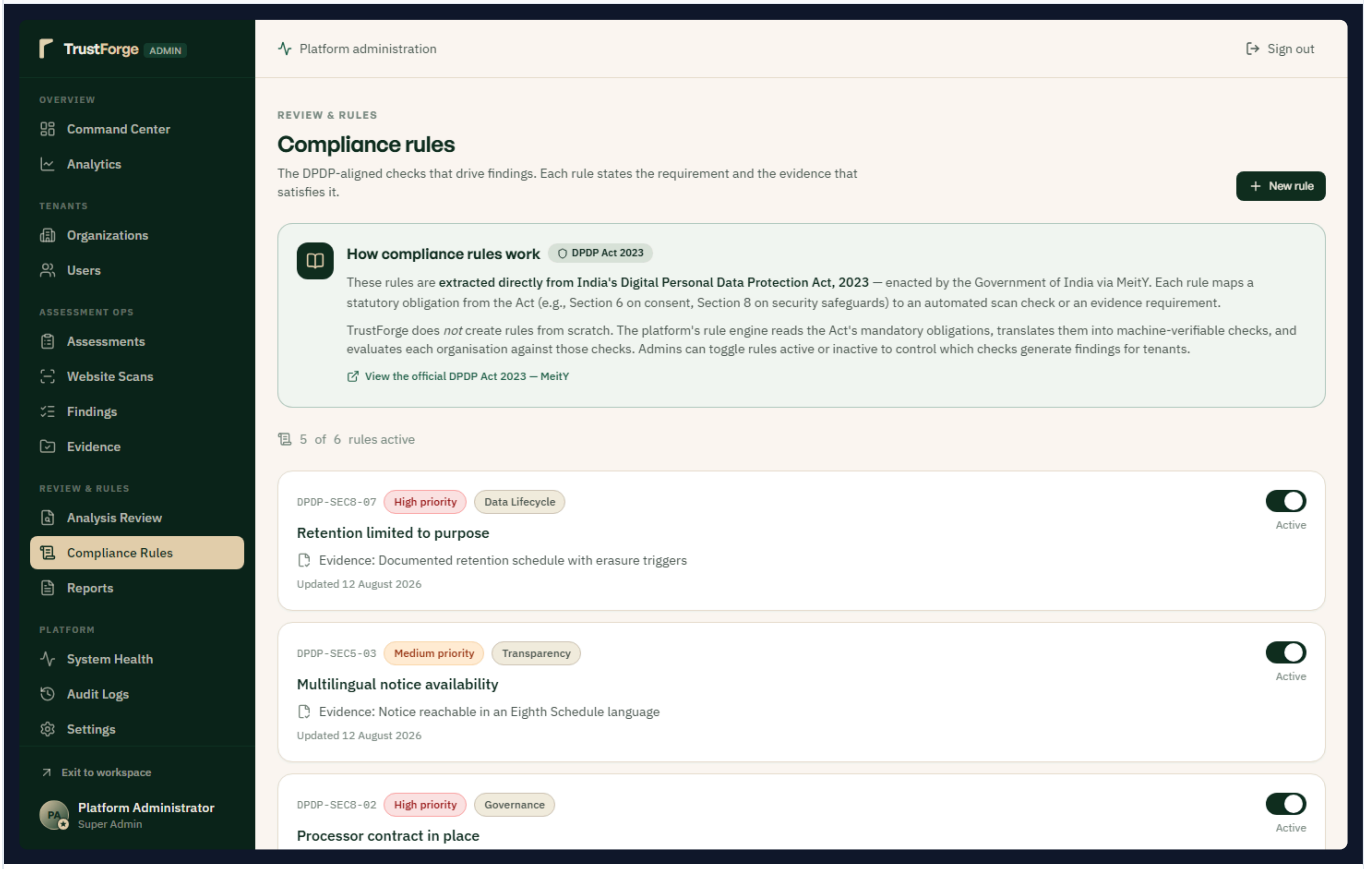
`Route: /admin/reviews` Role: Senior Certification Officer



Provides an administrative interface where compliance reviewers evaluate and certify completed tenant assessment submissions. Allows the reviewer to approve, request revisions on, or reject submitted assessments with itemized feedback. Approved reviews record an administrative certification timestamp and update the organization's official audit record.

Figure 46: Statutory Rules Engine & Legal Weight Tuning Matrix

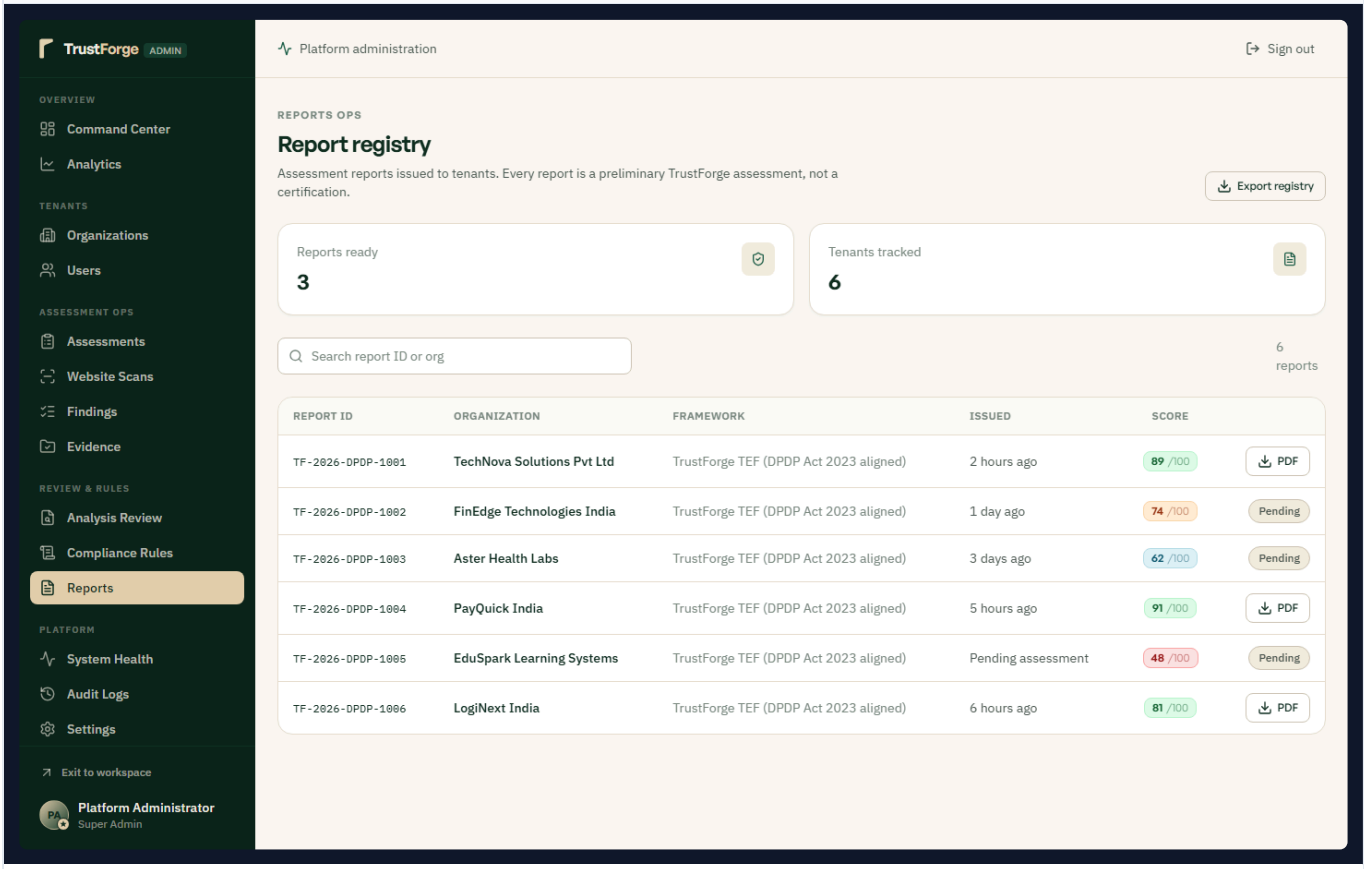
`Route: /admin/rules` Role: Legal / Rule Engineer



Allows authorized administrators to adjust statutory scoring formulas, section weights, and penalty parameters. Enables updates to assessment rule configurations as official notifications and subordinate rules are issued under the DPDP Act. Maintains versioned rule profiles to ensure audit assessments remain aligned with prevailing regulatory guidance.

Figure 47: Global Regulatory Export & Supervisory Report Vault

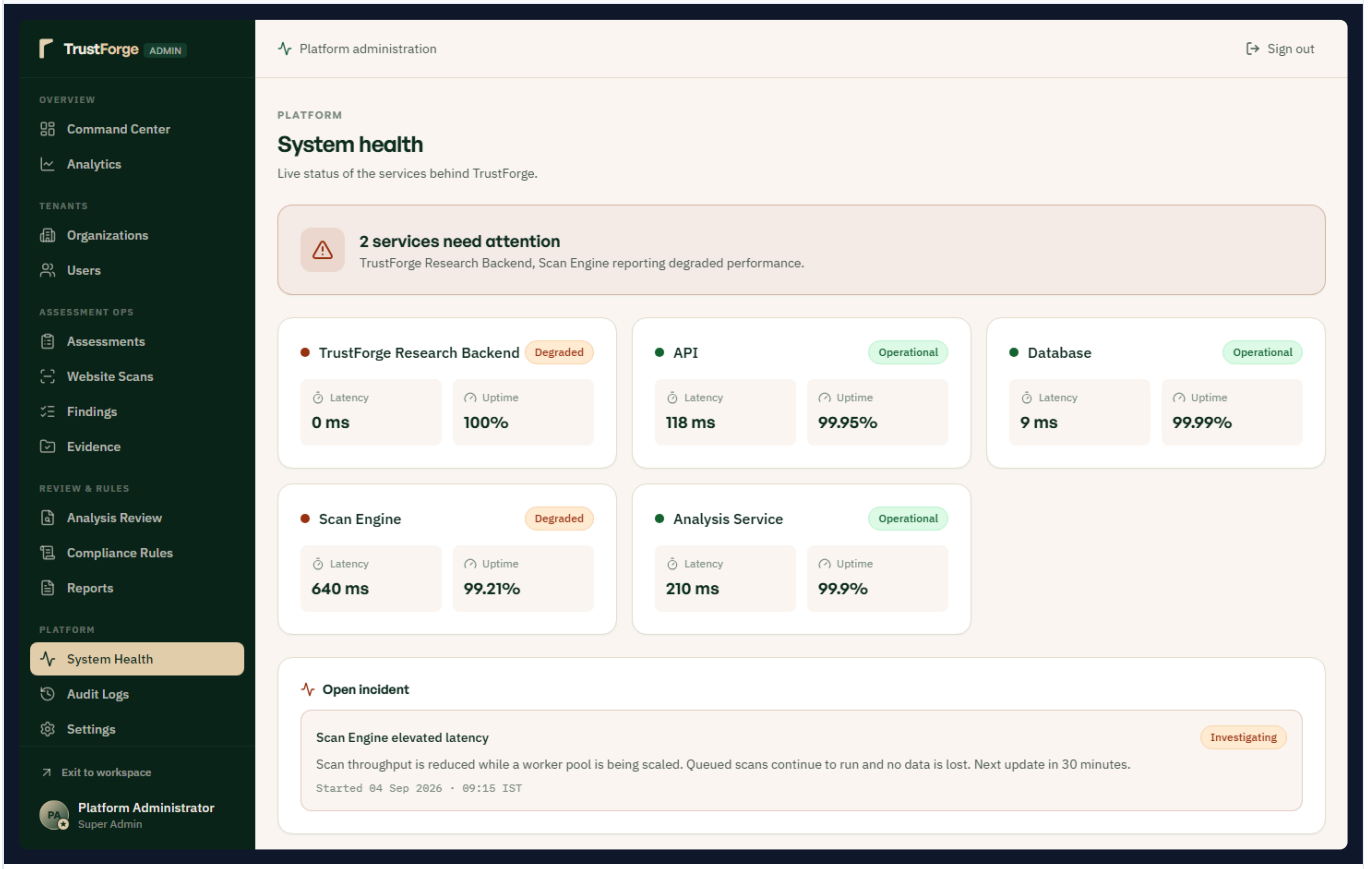
`Route: /admin/reports` Role: Supervisory Auditor



Maintains an archive of generated assessment summary reports, audit extracts, and regulatory disclosure packages. Enables administrators to view previously generated reports and review document generation timestamps. Maintains an audit record of report generation and download events for accountability purposes.

Figure 48: System Health Diagnostics & Infrastructure Monitoring

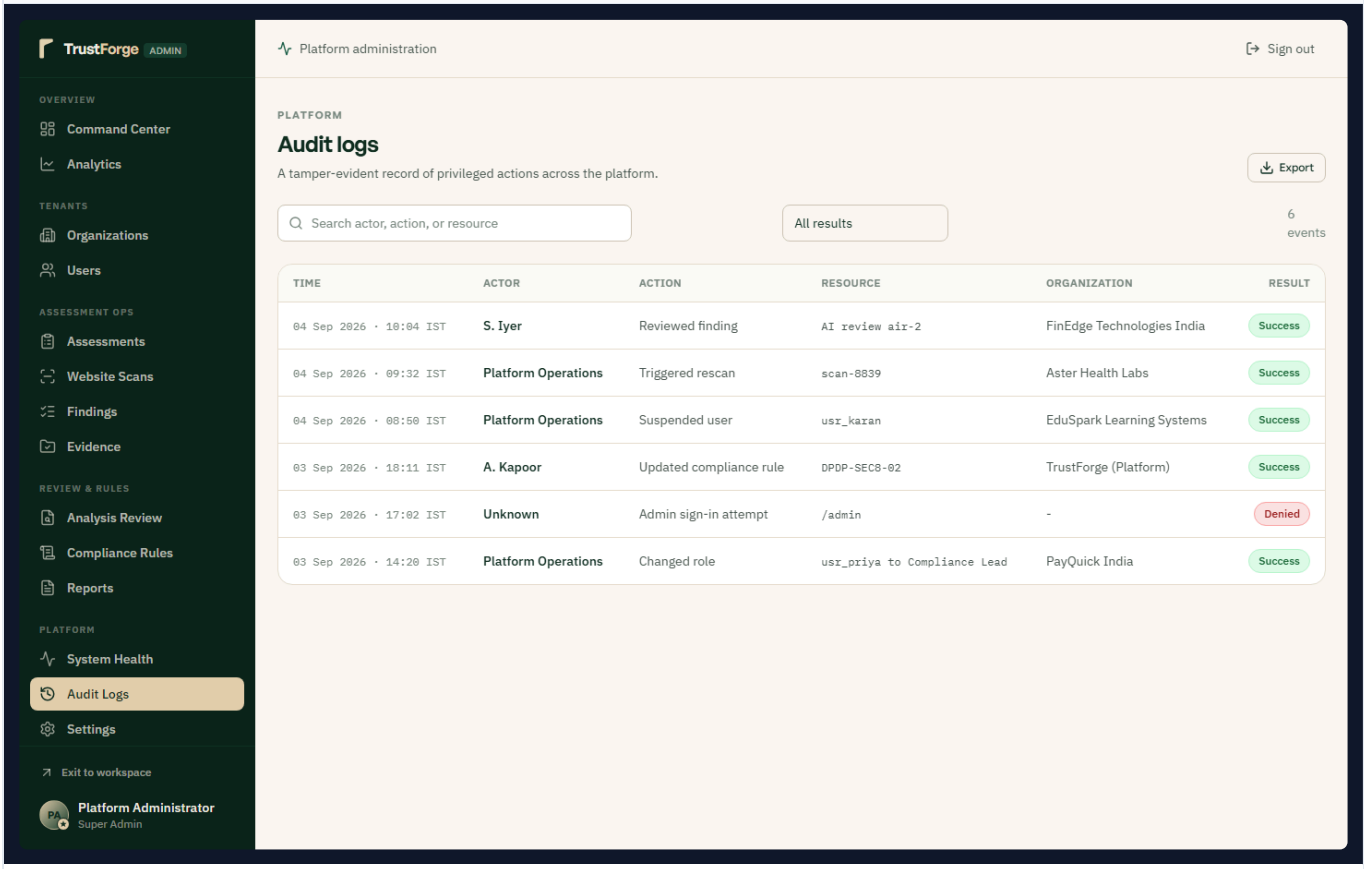
`Route: /admin/health` Role: DevOps / Infrastructure Lead



Monitors application service availability including Express API responsiveness, Python RAG worker status, and database state. Tracks memory usage, process uptime, active database connections, and background worker queue health. Provides diagnostic indicators to ensure system components operate within normal performance tolerances.

Figure 49: Immutable Security Audit Log & Forensic Activity Trail

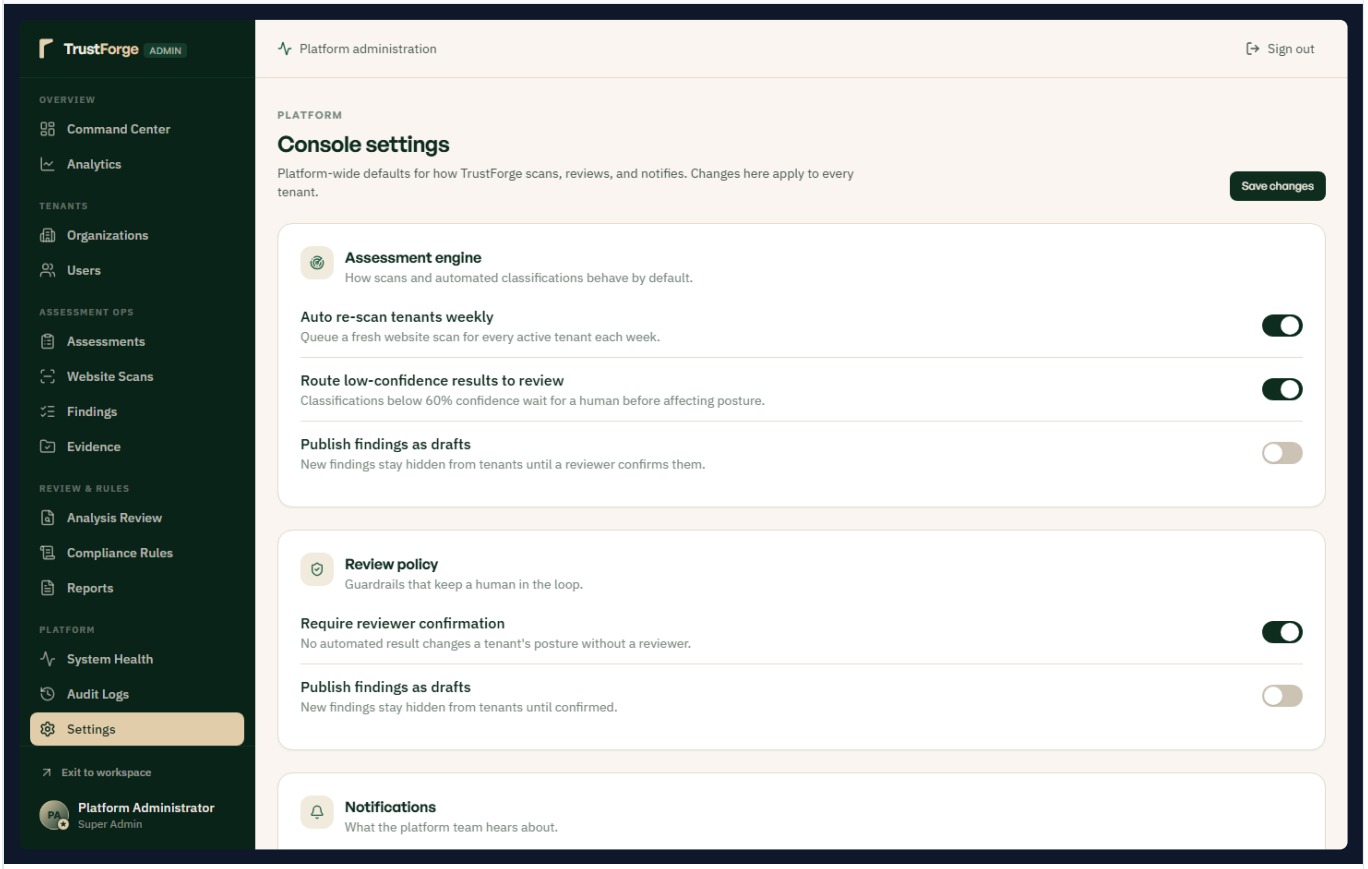
`Route: /admin/audit` Role: Forensic / Security Auditor



Maintains an append-only audit trail recording user logins, role changes, assessment submissions, and administrative updates. Each log entry records UTC timestamp, actor IP address, action performed, target resource, and result status. Provides an unalterable activity history for internal accountability and compliance verification.

Figure 50: Platform Global Settings & Policy Engine Configuration

`Route: /admin/settings` Role: Platform Superadmin



Configures platform security parameters including session timeout durations, CORS origin restrictions, and password policies. Manages automated maintenance schedules, logging verbosity, and zero-paid inference enforcement policies. Ensures consistent operational policies and security boundaries across the entire application deployment.